Acquisition Function Optimization Is NP-Hard, Even with Tuned Hyperparameters

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Abstract

We consider a zero-mean Gaussian process on $\mathbb{R}^D$ with an isotropic kernel $\sigma_f^2\, g(\|x-x'\|^2/\ell^2)$, where g is completely monotone, so the kernel is a scale mixture of Gaussians. We assume that the mixing measure of g has a finite moment of order slightly more than two, and that g decays at least polynomially and has a monotonicity property used to identify the lengthscale; in the noise-free model we also assume a mild tail or small-scale condition. We prove the monotonicity property for every standard family listed next, via an elasticity criterion and a Bessel-ratio inequality. This class contains the squared-exponential kernel, every Matérn kernel with $\nu > 2$ (so $\nu = \frac{5}{2}, \frac{7}{2}, \dots$), and the rational quadratic kernel (the generalized Cauchy kernel with exponent 2). We first prove that, with the hyperparameters supplied, approximating the minimum of the posterior mean over $[-1, 1]^D$ to within an additive constant is NP-hard in the strong sense. We then prove the same when the data are standardized and the length scale, signal variance and noise variance are set to their joint maximum-likelihood estimate. The hardness is not an artifact of the posterior mean alone: it persists for the lower confidence bound (with $\beta^{1/2}$ up to 59, and any polynomially bounded exploration weight after amplifying the gadget), expected improvement over a supplied incumbent above -62.5 (in standardized units) and, in the noise-free model, over the best observed value, probability of improvement over a supplied incumbent between -63 and -1.25, and the uniform-cost Gittins index (for every cost of polynomial bit length, after amplifying), via a single mechanism common to all of them. Finally, with the hyperparameters supplied, the problem in the noisy model is solvable in polynomial time when D is fixed and the ratio of the posterior mean's RKHS norm to the length scale is polynomially bounded, so, unless P = NP, the dimension of hard instances must grow.

The Max-Cut gadget is built by *moment matching*. Its weights make the posterior mean vanish to second order at every Gaussian scale, and the second-order term is exactly a shifted cut polynomial. Small scales are then controlled by a Taylor bound, and large scales by the tail of the mixing measure. A calibration design with exact invariant subspaces pins every joint MLE.

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1 Background and existing literature

Bayesian optimization has long been a useful way to optimize functions which are costly to evaluate. Using a Gaussian process prior, even a few observations can give enough information to change one's beliefs greatly. The primary question for the practitioner is to decide where to evaluate the function next after making some observations. A number of acquisition functions exist to represent how "promising" each potential point is as a possible improvement upon the data the user already has. In recent years, some commonly used ones have been the following:

- Expected improvement: The expected positive part of the improvement over the current running minimum at each point. If only one more observation will be made and the best observed value is reported, this is optimal; otherwise, we can view it as a greedy approximate dynamic programming algorithm (Garnett, 2023).
- Probability of improvement: The probability of the improvement over the current running minimum at each point. This function is often criticized for putting too much emphasis on exploitation as opposed to exploration (Garnett, 2023).
- Lower confidence bound: The infimum of a posterior credible interval of the function value at each point. This comes from the multi-armed bandits literature (Srinivas et al., 2010).
- Gittins index: Tailored to cases where there is a cost to make each observation rather than a fixed observation budget, the Gittins index represents gains relative to cost and includes the option for the user to decline to make an observation if no points are worth the cost. Introduced by Gittins (1979), followed up on in Gittins et al. (2011), and introduced to Bayesian optimization in Xie et al. (2024).

More details on acquisition functions can be found in Garnett (2023). Surprisingly little attention has been given to the question of how one would actually optimize these functions. Wilson et al. (2018) study gradient-based maximization of Monte Carlo acquisition functions and show that greedy batch construction is near-optimal for a submodular class. Schweidtmann et al. (2021) solve problems with Gaussian processes embedded by deterministic spatial branch-and-bound. More recently, Tang et al. (2026) offers a promising surrogate-based branch-and-bound approach, and

Xie et al. (2025) proposes a mixed-integer quadratic programming formulation, both of which would take exponential time in the worst case. To our knowledge, the question of whether a polynomial-time algorithm could succeed globally has not been addressed formally in the literature.

The closest complexity result we know of concerns kernel density estimates. Lee et al. (2021) show that finding an approximate mode of a Gaussian kernel density estimate is NP-hard in high dimension, and Luo et al. (2023) study dimensionality reduction for the same problem. A posterior mean with the squared-exponential kernel is also a sum of Gaussian bumps, but its weights $\Sigma^{-1}y$ are signed and are determined by the data. A reduction must therefore control these weights. It must also cover scale mixtures such as the Matérn kernels, and it must survive the fitting of the hyperparameters.

We prove here that for the most commonly used covariance kernels, the problem is NP-hard, meaning that there can be no polynomial-time algorithm to solve the problem unless P = NP. We do this via a reduction from Max-Cut, already known to be NP-hard (Karp, 1972; Garey et al., 1976). The proof follows the classical route from Max-Cut to indefinite quadratic minimization over a box, where multilinearity places a minimizer at a vertex (Sahni, 1974); Pardalos and Vavasis (1991) show that even one negative eigenvalue suffices.

Theorem 2 covers every computable kernel, in the sense of (A4), whose mixing measure has a finite moment of order slightly more than two. Theorem 3 adds the mild conditions (A2) and (A3) of Assumption 1, and in the noise-free model also (A5). This includes the squared-exponential kernel and every Matérn kernel with $\nu > 2$, such as the common default $\nu = \frac{5}{2}$, but not the Matérn kernels with $\nu \leq 2$, such as $\nu = \frac{3}{2}$ and $\nu = \frac{1}{2}$, for which $m_2 = \infty$. Whether the problem remains hard for these kernels is open.

Proof idea. The gadget is a signed combination of Gaussian bumps at $n_0 = 1 + 4m + 4d$ points, built by *moment matching*. Its weights make the combination vanish to second order in the Gaussian scale τ, and the second-order term is exactly a shifted cut polynomial, which is multilinear and therefore minimized at a vertex. For a scale mixture of Gaussians, small scales are controlled by a Taylor bound and large scales by the tail of the mixing measure, which is where the moment of order $2 + \omega$ enters. With the hyperparameters supplied, this already gives Theorem 2. To let the data choose the hyperparameters, we add calibration data whose likelihood splits exactly along invariant subspaces of the covariance matrix. Each subspace contributes a coercive term that pins one combination of the hyperparameters, and together they force every joint MLE into a narrow window around the target. The calibration points sit in blocks, separated from each other and from the gadget by many *hub* coordinates, so that far-away blocks are negligible; by Remark 40, exact invariant subspaces of this kind cost at least one dimension per block. Pairs of points at two spacings identify the length scale through the ratio in (A3). In the noisy model, replicated observations pin the noise variance, and in the noise-free model, squares of four points block the escape $\ell \to \infty$.

The regime of the hard instances. The hard instances are not typical Bayesian-optimization data. The gadget uses a length scale far larger than the box and weights of polynomial size, and in yes instances the posterior mean dips more than 60 fitted prior standard deviations below the prior mean. The likelihood tolerates this because the calibration data, almost all of the observations, set the hyperparameters. Remark 10 explains why the large length scale is what makes the gadget work, and Section 10 discusses how far these instances are from typical data.

Results and outline. Section 2 states the setting and the main theorems.

- *Fixed hyperparameters.* Theorem 2 shows that, with the hyperparameters supplied, approximating the minimum of the posterior mean to within an additive constant is NP-hard in the strong sense. Section 3 builds the moment-matched gadget and proves it, together with the corresponding hardness of ε-optimal values and points (Corollary 9). In the noise-free model the proof uses a lower bound on the smallest eigenvalue of the gadget Gram matrix (Corollary 50), which is proved in Appendix B with no further assumptions.
- *Fitted hyperparameters.* Theorem 3 shows the same when the data are standardized and every hyperparameter is set by maximum likelihood. Its proof is a chain of four steps.
  1. Section 4 builds one calibration design for both models, with exact invariant subspaces that split the likelihood into one term per group.
  2. Section 5 proves an abstract pinning proposition (Proposition 14) from three hypotheses (H1)–(H3), and verifies the identifiability hypothesis (H3). The same argument covers MAP estimates (Proposition 15).
  3. Section 6 checks (H1) and (H2) in the noisy model. Appendix B checks them in the noise-free model, where the spectral floor of the covariance matrix must be proved (Lemma 45).
  4. Section 7 bounds the posterior mean at every point of the resulting window and completes the proof. Appendix C bounds the contribution of the calibration data.
- *Acquisition functions.* Section 8 transfers hardness to every acquisition function that is monotone in the posterior mean and whose dependence on the posterior standard deviation is smaller than the gap (Lemma 23), including the lower confidence bound, expected improvement, probability of improvement and the Gittins index (Corollary 26). In the noise-free model this includes expected improvement over the best observed value (Proposition 28), and amplifying the gadget covers every polynomially bounded exploration weight and every Gittins cost of polynomial bit length (Proposition 29 and Lemma 25).
- *ε-optimality.* Section 8 also shows that computing ε-optimal values and points is hard, with supplied or fitted hyperparameters (Corollaries 32 and 33). With fitted hyperparameters, the reduction checks an output against the target without computing the MLE, using the stability of the posterior mean across the window (Lemma 31).
- *Dimension.* Section 9 shows that with the hyperparameters supplied, fixed dimension is easy in the noisy model when $\hat{H}/\ell$ is polynomially bounded, as it is on our instances, so, unless P = NP, the dimension of hard instances must grow.

Section 10 discusses limitations and open problems. Appendix A verifies the identifiability assumption (A3) for the standard kernel families.

2 Setting and main results

Kernels. Let $g : [0, \infty) \to (0, 1]$ be completely monotone with $g(0) = 1$. By Bernstein's theorem, there exists a probability measure π with $g(y) = \int_0^\infty e^{-ty} \, \mathrm{d}\pi(t)$. Define $m_k = \int t^k \, \mathrm{d}\pi$ and $F = 1 - g$. The kernel is $k_\ell(x, x') = g(\|x - x'\|^2/\ell^2)$, so

$$k_\ell(x, x') = \int e^{-\tau \|x - x'\|^2} \, \mathrm{d}\pi_\ell(\tau), \tag{1}$$

where π_ℓ is the image of π under $t \mapsto t/\ell^2$. By Schoenberg's theorem (Schoenberg, 1938), k_ℓ is positive semidefinite, and strictly positive definite on distinct inputs because $\pi \neq \delta_0$.

Assumption 1.

(A1) $m_{2+\omega} < \infty$ for some $\omega \in (0, 1]$, and $\pi \neq \delta_0$. Hence $0 < m_1, m_2 < \infty$.

(A2) $g(y) \leq C_g y^{-\vartheta}$ for all $y \geq 1$, for some $C_g, \vartheta > 0$.

(A3) $\mathcal{R}(v) := F(4v)/F(v)$ is strictly decreasing on $\mathbb{R}_+$.

(A4) $g(y)$ can be computed to b bits in time polynomial in b and the bit length of rational y.

(A5) Either (i) $g(u) \leq C_g \exp(-c_g u^{\vartheta'})$ for $u \geq 1$, or (ii) $\pi([u, 2u]) \geq c_\pi u^{b_\pi}$ for $0 < u \leq u_0$, for some $c_\pi, u_0 > 0$ and $b_\pi \geq 0$.

(A5) is used only in the noise-free model. The squared-exponential kernel ($\pi = \delta_{1/2}$) and the Matérn kernels satisfy (A5)(i), with $\vartheta' = 1$ and $\vartheta' = \frac{1}{2}$. For the rational quadratic kernel $g(y) = (1+y/c)^{-\alpha}$ (with the usual choice $c = 2\alpha$; the case $c = 1$ is the generalized Cauchy kernel $(1 + r^\varsigma)^{-\beta/\varsigma}$ with $\varsigma = 2$), π is a Gamma law, so all moments are finite, g decays polynomially, and (A5)(ii) holds. Generalized Cauchy kernels with $\varsigma < 2$ have $m_1 = \infty$ and are not covered. For Matérn with smoothness ν, π is the law of $\nu/(2S')$ with $S' \sim \text{Gamma}(\nu, 1)$, so $m_k < \infty$ if and only if $k < \nu$, and (A1) holds if and only if $\nu > 2$. Appendix A verifies (A3) for all of these families, via an elasticity criterion. (A5)(i) implies (A2), so (A2) is an extra assumption only for kernels covered by (A5)(ii), and those satisfy it. (A2) is used to make the separation S of Section 4 polynomial; the proof of Lemma 45 also uses its consequence $g(y) \to 0$, that is, that π has no atom at 0.

The reductions below are built for one fixed g. They use a few constants that depend only on g, such as m_1, m_2, $m_{2+\omega}$, and the quantities v_1, T_1, t_1, p_1, t_{med} introduced later. Some of these are only shown to exist, but each is a fixed number, so rational bounds on them can be hard-coded into the reduction.

Model. The prior is $\sigma_f^2 k_\ell$ with zero mean. In the *noisy model*, the observations carry i.i.d. $\mathcal{N}(0, \sigma_n^2)$ noise, and $\theta = (\ell^2, \sigma_f^2, \sigma_n^2)$. In the *noise-free model*, $\sigma_n = 0$, the inputs are distinct, and $\theta = (\ell^2, \sigma_f^2)$. Define $\Sigma_\theta = \sigma_f^2 K_\ell + \sigma_n^2 I$. Up to an additive constant,

$$L(\theta) = -\frac{1}{2}\big[\log\det\Sigma_\theta + y^\top \Sigma_\theta^{-1} y\big], \qquad \mu_\theta(x) = \sigma_f^2\, k_\ell(x)^\top \Sigma_\theta^{-1} y.$$

A joint MLE is any maximizer of L.

Main results. The first result fixes the hyperparameters. It isolates the gadget, and it needs neither standardization nor (A2), (A3) and (A5).

Theorem 2 (Fixed hyperparameters)**.** *Assume (A1) and (A4). In both the noisy and the noise-free model, the following promise problem is NP-hard in the strong sense. The input is a dimension D, points $x_1, \ldots, x_n \in [-1, 1]^D$, observations $y \in \mathbb{Q}^n$, and hyperparameters $\ell^2 \in \mathbb{Q}_{>0}$, $\sigma_f^2 = 1$, and (if applicable) $\sigma_n^2 \in \mathbb{Q}_{>0}$. Decide whether*

$$\min_{[-1,1]^D} \mu_\theta \leq -95.9 \qquad \textit{or} \qquad \min_{[-1,1]^D} \mu_\theta \geq -0.05.$$

Because $\sigma_f^2 = 1$, the thresholds are in units of the prior standard deviation. Here and below, *NP-hard in the strong sense* means that the problem remains NP-hard when every number in the input is a rational whose numerator and denominator are bounded by a polynomial in the input size. Kernel values never appear in the input; the reduction computes them internally using (A4). The second result lets the data choose the hyperparameters.

Theorem 3 (Fitted hyperparameters)**.** *Assume (A1)–(A4) in the noisy model, or (A1)–(A5) in the noise-free model. The following promise problem is NP-hard in the strong sense. The input is a dimension D, points $x_1, \dots, x_n \in [-1,1]^D$, and observations $y \in \mathbb{Q}^n$ with*

$$\frac{1}{n}\sum_i y_i = 0, \qquad \frac{1}{n}\sum_i y_i^2 \in \Big[1 - \frac{1}{n}, 1\Big].$$

The promise is that either

$$\min_{[-1,1]^D} \mu_{\hat\theta} \le -c_{\text{yes}}, \quad c_{\text{yes}} := 63, \quad \textit{for every joint MLE } \hat\theta$$

or

$$\min_{[-1,1]^D} \mu_{\hat\theta} \ge -c_{\text{no}}, \quad c_{\text{no}} := 1.25, \quad \textit{for every joint MLE } \hat\theta;$$

decide which. On every instance produced by the reduction, the promise holds, a joint MLE exists, and every joint MLE lies within relative distance $1/\mathrm{poly}$ *of an explicit point $\theta^\circ_{\mathrm{s}}$, the target θ° rescaled by the standardizing factor.*

Lemma 4 (Scale equivariance)**.** *Let $c_y > 0$. Replacing y by $c_y y$ multiplies σ_f^2 and σ_n^2 at every joint MLE by c_y^2, leaves ℓ unchanged, and multiplies μ by c_y.*

Proof. With $\theta_c = (\ell^2, c_y^2\sigma_f^2, c_y^2\sigma_n^2)$, we have $\Sigma_{\theta_c} = c_y^2\Sigma_\theta$, so $L(\theta_c; c_y y) = L(\theta; y) - n\log c_y$ and $\mu_{\theta_c}(\cdot; c_y y) = c_y\,\mu_\theta(\cdot; y)$. □

Standardization is what makes the additive thresholds of Theorem 3 meaningful. Without it, by Lemma 4, rescaling y would turn any additive gap into any other.

3 The moment-matched gadget

An instance is a graph $G = ([d], E)$ with $|E| = m$ and an integer threshold $c_\star$. A cut is a vertex $v \in \{\pm 1\}^d$, and

$$\mathrm{cut}(v) = |\{\{j,k\} \in E : v_j \ne v_k\}|.$$

The instance is a *yes* instance if $\mathrm{maxcut}(G) \ge c_\star + 1$ and a *no* instance if $\mathrm{maxcut}(G) \le c_\star$; deciding this is NP-complete (Karp, 1972; Garey et al., 1976). Since $\mathrm{maxcut} \ge \lceil m/2 \rceil$, instances with $c_\star < m/2$ are yes instances and those with $c_\star \ge m$ are no instances; we map both to fixed instances with the same answer. Padding with isolated vertices, we assume $m \ge 1$ and $d \ge 5$. This implies $m/2 \le c_\star \le m - 1$, and we set $q = 2c_\star - m \in [0, m-2]$ and $\beta_0 = 32(q + \frac{1}{2})$.

We may also assume $q \ge 1$. Adding a disjoint edge on two new vertices raises both m and $\mathrm{maxcut}(G)$ by 1. Raising $c_\star$ by 1 as well preserves the answer and the range of $c_\star$, and it raises q by 1. This normalization is used only in Lemma 27 of Section 8.

Points and weights.

location	weight	count
0	$4\beta_0$	1
$ae_j + be_k$, $\{j,k\} \in E$, $a,b \in \{\pm 1\}$	ab	$4m$
$\pm e_i$, $i \in [d]$	$\beta_{\text{out}} = 2\beta_0/(3d)$	$2d$
$\pm \frac{1}{2} e_i$, $i \in [d]$	$\beta_{\text{in}} = -8\beta_0/(3d)$	$2d$

These are $n_0 = 1 + 4m + 4d$ distinct points P with weight vector α^*, and

$$\|\alpha^*\|_1 = 4m + 4\beta_0 + \frac{20}{3}\beta_0 \le 360(m+1).$$

Lemma 5 (Moments)**.** *Let*

$$\mathcal{Q} = \sum_p \alpha_p^* pp^\top, \quad C(x) = \sum_{\{j,k\}\in E} x_j x_k.$$

Then

$$\sum_p \alpha_p^* = 0, \quad \sum_p \alpha_p^* p = 0, \quad \sum_p \alpha_p^* \|p\|^2 = 0, \quad x^\top \mathcal{Q} x = 8C(x), \quad \sum_p \alpha_p^* \|p\|^4 = \beta_0.$$

Proof. Each edge quadruple $\{ae_j + be_k\}$ has weights summing to 0, first moment

$$\sum_{a,b} ab(ae_j + be_k) = 0,$$

constant $\|p\|^2 = 2$, and second moment

$$\sum_{a,b} ab(ae_j + be_k)(ae_j + be_k)^\top = 4(e_j e_k^\top + e_k e_j^\top).$$

The axis families are symmetric, so their first moment is 0. They contribute

- $2d(\beta_{\text{out}} + \beta_{\text{in}})$ to the total weight,
- $2d(\beta_{\text{out}} + \frac{1}{4}\beta_{\text{in}}) = 0$ to $\sum \alpha\|p\|^2$,
- the matrix $2(\beta_{\text{out}} + \frac{1}{4}\beta_{\text{in}})I = 0$ to $\mathcal{Q}$, and
- $2d(\beta_{\text{out}} + \frac{1}{16}\beta_{\text{in}}) = 2d\beta_0(\frac{2}{3d} - \frac{1}{6d}) = \beta_0$ to $\sum \alpha\|p\|^4$.

Finally, $4\beta_0 + 2d(\beta_{\text{out}} + \beta_{\text{in}}) = 4\beta_0 - 4\beta_0 = 0$. □

Lemma 6 (Second-order structure)**.** *Let*

$$G_\tau(x) = \sum_p \alpha_p^* e^{-\tau\|x-p\|^2}, \quad \Lambda_E = \beta_0 + 32C.$$

For $x \in [-1,1]^d$, the expansion of G_τ in τ has zero coefficients at orders 0 and 1, and

$$G_\tau(x) = \frac{\tau^2}{2}\Lambda_E(x) + R_\tau(x), \qquad |R_\tau(x)| \le \frac{\tau^3}{6}\|\alpha^*\|_1 \max_p \|x-p\|^6 \le \frac{\tau^3}{6}\|\alpha^*\|_1 (4d)^3.$$

Λ_E is multilinear, and at a vertex,

$$\Lambda_E(v) = 32\left(2\big(c_\star - \text{cut}(v)\big) + \frac{1}{2}\right).$$

Hence $\Lambda_E \ge 16$ on $[-1,1]^d$ in no instances, and $\Lambda_E(v^*) \le -48$ at every maximum-cut vertex v^* in yes instances.

Proof. Let $u_p = \tau\|x-p\|^2 \in [0, 4d\tau]$. Here, $\|x-p\|^2 \le d+6 \le 4d$, because p has at most two nonzero entries, each of size at most 1. The bound

$$\left| e^{-u} - 1 + u - \frac{u^2}{2} \right| \le \frac{u^3}{6} \quad \text{for} \quad u \ge 0$$

gives $G_\tau = \sum \alpha_p^*(1-u_p+\frac{1}{2}u_p^2)+R_\tau$ with the stated bound. The orders 0 and 1 vanish by Lemma 5. Expand

$$\|x-p\|^4 = (\|x\|^2+\|p\|^2)^2 - 4(\|x\|^2+\|p\|^2)\, x\cdot p + 4(x\cdot p)^2.$$

The first term sums to β_0 by Lemma 5. The second sums to 0, because $\sum \alpha^* p = 0$ and $\|p\|^2(x\cdot p)$ is odd within each symmetric family while $\|p\|^2 = 2$ is constant on edge groups. The third gives $4x^\top \mathcal{Q} x = 32C$. The order-2 coefficient is therefore $\frac{1}{2}\Lambda_E$. It is affine in each coordinate, so its minimum over $[-1,1]^d$ is attained at a vertex, where

$$C(v) = m - 2\,\mathrm{cut}(v), \qquad \Lambda_E(v) = 32\left(q + \frac{1}{2} + m - 2\,\mathrm{cut}(v)\right). \qquad \square$$

Lemma 7 (Kernel facts)**.** *For $u, v \ge 0$:*

(a) $|g'(u)| \le m_1$, $|g''(u)| \le m_2$, $F(v) \le m_1 v$, *and F is concave and increasing.*

(b) $F(v) \ge \frac{1}{2}m_1 v$ *whenever* $v \le m_1/m_2$.

(c) $|ug'(u)| = \int tu\, e^{-tu}\, \mathrm{d}\pi \le e^{-1}$.

(d) $\int (1-e^{-tv})^2\, \mathrm{d}\pi(t) \le m_2 v^2$, *and* $\int (1-e^{-tv})^2\, \mathrm{d}\pi(t) \ge \frac{1}{8}m_2 v^2$ *whenever* $\int_{t>1/v} t^2\, \mathrm{d}\pi \le \frac{1}{2}m_2$.

(e) *For every squared distance $u \ge 0$ and every ℓ^2 with $|\ell^2/\Lambda - 1| \le \eta \le \frac{1}{50}$,*

$$\left|g(u/\ell^2) - g(u/\Lambda)\right| \le 0.38\,\eta \quad \textit{and} \quad \left|g(u/\ell^2) - g(u/\Lambda)\right| \le 0.76\,\eta\, g\left(\frac{u}{3\Lambda}\right).$$

Proof. We will prove each part individually.

(a) Differentiate under the integral and use $1-e^{-x} \le x$. $F' = -g' > 0$ is decreasing because g is convex.

(b) Use $1-e^{-x} \ge x - \frac{1}{2}x^2$, so $F(v) \ge m_1 v - \frac{1}{2}m_2 v^2 \ge \frac{1}{2}m_1 v$.

(c) $xe^{-x} \le e^{-1}$.

(d) The upper bound uses $(1-e^{-x})^2 \le x^2$. For the lower bound, $1-e^{-x} \ge x/2$ for $x \le 1$, so the integral is at least $\frac{v^2}{4}\int_{t\le 1/v} t^2\, \mathrm{d}\pi \ge \frac{v^2}{4}\cdot\frac{m_2}{2}$.

(e) Let $\chi(\lambda) = g(\lambda u)$. By (c),

$$|\chi'(\lambda)| = |u'g'(u')|/\lambda \le 1/(e\lambda)$$

with $u' = \lambda u$. Integrating $\mathrm{d}\lambda/\lambda$ between $1/\ell^2$ and $1/\Lambda$ gives a factor

$$|\log(\Lambda/\ell^2)| \le \eta/(1-\eta) \le 1.03\eta,$$

and $1.03/e \le 0.38$. For the second bound, use $xe^{-x} \le \frac{2}{e}e^{-x/2}$:

$$|\chi'(\lambda)| = \lambda^{-1}\int tu\lambda\, e^{-t\lambda u}\, \mathrm{d}\pi \le \frac{2}{e\lambda}\, g(\lambda u/2).$$

For λ between $1/\ell^2$ and $1/\Lambda$ we have

$$\lambda u/2 \geq u/(2.04\Lambda) \geq u/(3\Lambda),$$

and g is decreasing, so integrating as for the first bound gives the factor $\frac{2}{e} \cdot 1.03 \leq 0.76$. □

The first bound in (e) does not depend on the distance, which matters because points of the box can be far from the gadget in the auxiliary coordinates. The second is needed when many far-apart pairs are summed, as in the family-sum block of Lemma 19.

Proposition 8 (Margin over a window)**.** *Let*

$$\tau_0 = 3/(4d^3\|\alpha^*\|_1), \quad \mu^*_\ell(x) = \sum_p \alpha^*_p k_\ell(x,p) = \int G_\tau(x)\,\mathrm{d}\pi_\ell(\tau).$$

For a length scale ℓ, let

$$Q_\ell = \int_{(0,\tau_0]} \tau^2\,\mathrm{d}\pi_\ell, \quad R_\ell = \frac{8}{\tau_0}\int_{(0,\tau_0]} \tau^3\,\mathrm{d}\pi_\ell, \quad E_\ell = \|\alpha^*\|_1\,\pi_\ell((\tau_0,\infty)).$$

(i) *In yes instances, $\mu^*_\ell(v^*) \leq -24Q_\ell + R_\ell + E_\ell$.*

(ii) *In no instances, $\int e^{-\tau t}G_\tau(x)\,\mathrm{d}\pi_\ell(\tau) \geq -E_\ell$ for every $x \in [-1,1]^d$ and $t \geq 0$.*

Let $\varepsilon_{\mathrm{m}} \in (0, \frac{1}{2}]$, and let

$$K_\omega := \Big(32 + \frac{\|\alpha^*\|_1}{\tau_0^2}\Big)\frac{m_{2+\omega}}{m_2}, \qquad T_\star := \Big(\frac{K_\omega}{\varepsilon_{\mathrm{m}}}\Big)^{1/\omega}.$$

If $\Lambda \geq 2T_\star/\tau_0$, then for every $\ell^2 \in [\Lambda/2, 2\Lambda]$,

$$Q_\ell \geq (1-\varepsilon_{\mathrm{m}})\frac{m_2}{\ell^4}, \qquad -24Q_\ell + R_\ell + E_\ell \leq -(24-\varepsilon_{\mathrm{m}})\frac{m_2}{\ell^4}, \qquad E_\ell \leq \varepsilon_{\mathrm{m}}\frac{m_2}{\ell^4}.$$

Consequently, with $\gamma := m_2/\Lambda^2$, $\varepsilon_{\mathrm{m}} = 0.01$ and $|\ell^2/\Lambda - 1| \leq 0.02$: in yes instances $\mu^*_\ell(v^*) \leq -23.05\,\gamma$, and in no instances the integral in (ii) is at least $-0.011\,\gamma$. At $\ell^2 = \Lambda$ these become $-23.99\,\gamma$ and $-0.01\,\gamma$. A subscript Λ, as in μ^*_Λ, Q_Λ and k_Λ, always means the value at $\ell^2 = \Lambda$.

Proof. For $\tau \leq \tau_0$, Lemma 6 gives

$$|R_\tau| \leq \frac{32}{3}d^3\|\alpha^*\|_1\tau^3,$$

and since $d^3\|\alpha^*\|_1 = 3/(4\tau_0)$,

$$|R_\tau| \leq 8\tau^2\frac{\tau}{\tau_0} \leq 8\tau^2.$$

So in no instances,

$$G_\tau \geq 8\tau^2 - 8\tau^2 = 0 \quad \text{on} \quad [-1,1]^d,$$

and in yes instances,

$$G_\tau(v^*) \leq -24\tau^2 + \frac{8\tau^3}{\tau_0}.$$

For $\tau > \tau_0$, every $e^{-\tau\|x-p\|^2} \in (0,1]$, so $|G_\tau| \leq \|\alpha^*\|_1$. Integrating gives (i) and (ii); the factor $e^{-\tau t} \in (0,1]$ preserves the sign at every scale $\tau \leq \tau_0$ and the bound at every scale $\tau > \tau_0$.

For the second claim, let $T = \tau_0 \ell^2 \geq \tau_0 \Lambda/2 \geq T_\star$, so that $T^\omega \geq K_\omega/\varepsilon_{\mathrm{m}}$. Since π_ℓ is the image of π under $t \mapsto t/\ell^2$, and $t^2 \leq t^{2+\omega} T^{-\omega}$ for $t > T$,

$$Q_\ell = \ell^{-4} \int_{t \leq T} t^2 \, \mathrm{d}\pi \geq \frac{m_2 - m_{2+\omega} T^{-\omega}}{\ell^4}.$$

Similarly, $t^3 \leq t^{2+\omega} T^{1-\omega}$ for $t \leq T$, so

$$R_\ell = \frac{8}{\tau_0 \ell^6} \int_{t \leq T} t^3 \, \mathrm{d}\pi \leq \frac{8 T^{1-\omega} m_{2+\omega}}{\tau_0 \ell^6} = \frac{8 m_{2+\omega} T^{-\omega}}{\ell^4}.$$

By Markov's inequality,

$$\pi_\ell((\tau_0, \infty)) = \pi(t > T) \leq m_{2+\omega} T^{-2-\omega},$$

and $T^2 = \tau_0^2 \ell^4$, so

$$E_\ell \leq \frac{\|\alpha^*\|_1 m_{2+\omega} T^{-\omega}}{\tau_0^2 \ell^4}.$$

Adding the three losses,

$$24\Big(\frac{m_2}{\ell^4} - Q_\ell\Big) + R_\ell + E_\ell \leq \Big(32 + \frac{\|\alpha^*\|_1}{\tau_0^2}\Big) \frac{m_{2+\omega} T^{-\omega}}{\ell^4} = \frac{K_\omega m_2 T^{-\omega}}{\ell^4} \leq \varepsilon_{\mathrm{m}} \frac{m_2}{\ell^4}.$$

This gives all three bounds, since each loss is nonnegative.
For the last claim, $|\ell^2/\Lambda - 1| \leq 0.02$ gives $0.9604\Lambda^2 \leq \ell^4 \leq 1.0404\Lambda^2$. In yes instances,

$$\mu_\ell^*(v^*) \leq -\frac{23.99\, m_2}{1.0404 \Lambda^2} \leq -23.05\gamma.$$

In no instances, the integral is at least

$$-E_\ell \geq -\frac{0.01\, m_2}{0.9604 \Lambda^2} \geq -0.011\gamma.$$

All of τ_0^{-1}, $T_\star$ and Λ are polynomial in d and $1/\varepsilon_{\mathrm{m}}$, with constants depending only on g. □

Proof of Theorem 2

Choose Λ as in Proposition 8 with $\varepsilon_{\mathrm{m}} = 0.01$, and let $\gamma = m_2/\Lambda^2$ and $M = 4/\gamma$. The inputs are the $n = n_0$ gadget points $P \subset [-1,1]^d$, so $D = d$.

Noisy model. Take $\theta = (\Lambda, 1, 1)$. Let y be $\Sigma_\theta M \alpha^*$, rounded to a multiple of $\rho := 0.01/n_0$. By (A4), the kernel entries can be computed accurately enough that every entry of the rounding error e satisfies $|e_p| \leq \rho$. The weights are

$$\Sigma_\theta^{-1} y = M\alpha^* + \Sigma_\theta^{-1} e, \qquad \|\Sigma_\theta^{-1} e\|_1 \leq \sqrt{n_0}\, \|\Sigma_\theta^{-1} e\|_2 \leq n_0 \rho = 0.01,$$

because $\lambda_{\min}(\Sigma_\theta) \geq \sigma_n^2 = 1$. Since $k_\Lambda \leq 1$, it follows that

$$|\mu_\theta - M\mu_\Lambda^*| \leq 0.01 \quad \text{on} \quad [-1,1]^d.$$

By Proposition 8 at $\ell^2 = \Lambda$, with $M\gamma = 4$,

- in yes instances, $\min \mu_\theta \leq \mu_\theta(v^*) \leq -4 \cdot 23.99 + 0.01 \leq -95.9$;
- in no instances, $\min \mu_\theta \geq -4 \cdot 0.01 - 0.01 = -0.05$.

Noise-free model. Fix $t_1 > 0$ with $p_1 := \pi([t_1, 2t_1]) > 0$. Take $\theta = (\Lambda, 1)$ and $y = K_{GG}(\Lambda)M\alpha^*$, rounded to a multiple of $\rho := 0.01\lambda_G^-/n_0$, where K_{GG} is the Gram matrix of P and

$$\lambda_G^- := c_G p_1 \min\left\{1, \left(\frac{t_1}{2\Lambda}\right)^4\right\}$$

is a lower bound on $\lambda_{\min}(K_{GG}(\ell))$ for every $\ell^2 \le 2\Lambda$, by Corollary 50 (proved in Appendix B). It is 1/poly, and Section 7 uses the same λ_G^-. The same argument, with $\lambda_{\min}(\Sigma_\theta) \ge \lambda_G^-$, gives the same bounds.

Size. The coordinates of the points lie in $\{0, \pm\frac{1}{2}, \pm 1\}$, Λ may be taken to be an integer, and every observation is a multiple of ρ with $|y_p| \le n_0(1 + M\|\alpha^*\|_1)$. All of these are polynomial in d, so every number in the instance has polynomially bounded numerator and denominator, and the reduction is strong. □

Corollary 9 (Approximation, fixed hyperparameters)**.** *Unless* P = NP*, no polynomial-time algorithm, given an instance as in Theorem 2, outputs a number within ε of* $\min \mu_\theta$ *for any $\varepsilon < 47.9$, or a rational $\hat{x}$ with $\mu_\theta(\hat{x}) \le \min \mu_\theta + \varepsilon$ for any $\varepsilon < 95.7$.*

Proof. For values, compare the output with -47.975: it is at most $-95.9 + \varepsilon < -47.975$ in yes instances and at least $-0.05 - \varepsilon > -47.975$ in no instances. For points, compute $\mu_\theta(\hat{x})$ to within 0.05 as in Lemma 30 of Section 8. That lemma is stated for the instances of Theorem 3, but its proof uses only (A4) and a 1/poly lower bound on $\lambda_{\min}(\Sigma_\theta)$; here $\lambda_{\min}(\Sigma_\theta)$ is at least 1 or λ_G^-. Then compare with -0.1: the result is at most $-95.9 + \varepsilon + 0.05 < -0.1$ in yes instances and at least -0.1 in no instances. □

Remark 10 (Why the gadget works)**.** Because the squared length scale Λ is polynomially large in d, far larger than the squared diameter $4d$ of the box, every kernel is close to its second-order Taylor expansion on the box. This is what the moment matching exploits.

4 One calibration template for both models

We now add calibration data that force every joint MLE into a window around the hyperparameters used in the proof of Theorem 2. Sections 4–7 prove Theorem 3.

Coordinates. We work in $\mathbb{R}^D$ with d main coordinates, a hub set S_b of size S for each block b (disjoint across blocks), and private coordinates for each block. All points equal -1 in every hub and private coordinate unless stated otherwise. Gadget observations sit at $(p, -1)$ for $p \in P$.

Families. Each family has B blocks, with B even. Block points have main coordinates 0 and hub coordinates $+1$ on S_b.

- *Pairs* P_h, $h \in \{1, 2\}$: block b has two points, and point $j \in \{1, 2\}$ sets its private coordinate $\iota_{b,j}$ to $-1 + h$.
- *Squares* Sq ($h = 1$, noise-free model only): block b has four private coordinates $\iota_{b,\pm}, \jmath_{b,\pm}$ and four points indexed by $(a, a') \in \{\pm 1\}^2$. Point (a, a') sets $\iota_{b,a}$ and $\jmath_{b,a'}$ to $-1 + h$.

Model	Families $\mathcal{F}$	Replicates	Role of the extra piece
Noisy	P_1, P_2	r (even)	replicate contrasts pin σ_n^2
Noise-free	P_1, P_2, Sq	1	squares block the escape $\ell \to \infty$

Let k_f be the block size of family f, $n_f = k_f Br$ be the number of observations in it, $n_{\max} = \max_f n_f$, and ϱ_f be the private squared norm of a block point after translating by $+1$ (h^2 for pairs, 2 for squares). The squared distances are as follows.

- Within a pair: $2h^2$.
- Within a square: 2 between points differing in one sign and 4 between opposite points.
- Between blocks of families f and f': $8S + \varrho_f + \varrho_{f'}$.
- From a gadget point $(p, -1)$ to a block of family f: $\|p\|^2 + 4S + \varrho_f$.

Every observation outside a block is equidistant from all points of that block, and every observation outside a family is equidistant from all points of that family.

Lemma 11 (Invariant subspaces)**.** *Let $g_\ell(y) = g(y/\ell^2)$ and define $\kappa_g(\ell)$ by:*

group	*dimension*	$\kappa_g(\ell)$
P_h *within:* $(+1, -1)$ *on a pair*	B	$1 - g_\ell(2h^2)$
P_h *between:* $\sum_b c_b \mathbf{1}_b$, $\sum c_b = 0$	$B - 1$	$1 + g_\ell(2h^2) - 2g_\ell(8S + 2h^2)$
Sq *pattern* a*, and pattern* a'	B *each*	$1 - g_\ell(4)$
Sq *pattern* aa'	B	$1 - 2g_\ell(2) + g_\ell(4)$
Sq *between:* $\sum_b c_b \mathbf{1}_b$, $\sum c_b = 0$	$B - 1$	$1 + 2g_\ell(2) + g_\ell(4) - 4g_\ell(8S + 4)$

Here "pattern a" is the vector with value a at point (a, a') of one square, and similarly for a' and aa'. In both models, each group vector, extended constantly across replicates, is an eigenvector of Σ_θ for every θ and every isotropic kernel, with eigenvalue

$$\lambda_g(\theta) = r\,\sigma_f^2\,\kappa_g(\ell) + \sigma_n^2.$$

In the noisy model the replicate contrasts (vectors supported on the replicates of one point and summing to zero there) form a further group of dimension $n_R = 4B(r-1)$ with eigenvalue σ_n^2. The complement W is spanned by the n_0 gadget coordinates and the normalized family sums $u_f = \mathbf{1}_f/\sqrt{n_f}$; it is invariant, and $\dim W = n_W = n_0 + |\mathcal{F}|$.

Proof. Replicates are identical inputs, so K_ℓ annihilates every replicate contrast and its eigenvalue is σ_n^2. For a vector v constant across replicates,

$$(K_\ell v)_{(x,k)} = r \sum_{x'} k_\ell(x, x') v_{x'},$$

so it suffices to compute the noise-free eigenvalues and multiply by r. Every listed vector sums to zero on each block (within and pattern groups) or on each family (between groups). A row of K_ℓ indexed by an observation outside that block or family is constant on it, so it annihilates the vector. For a pair, the row at the first point gives $1 - g_\ell(2h^2)$. For pattern aa', the row at (a, a') gives $aa'(1 - 2g_\ell(2) + g_\ell(4))$, since the two neighbors carry the opposite sign and the opposite corner the same sign; patterns a and a' are similar, with one neighbor of each sign. For the between groups, a row in block b gives c_b times the block's row sum plus

$$\sum_{b' \neq b} c_{b'} k_f g_\ell(8S + 2\varrho_f) = -c_b k_f g_\ell(8S + 2\varrho_f).$$

The complement of an invariant subspace of a symmetric matrix is invariant. □

Data. Let Λ be the target squared length scale, $\theta^\circ = (\Lambda, 1, 1)$ or $(\Lambda, 1)$, $\gamma = m_2/\Lambda^2$ and $M = 4/\gamma$. The data are a sum of independent pieces, one for each group and one on W:

- *Replicates* (noisy model): the k-th replicate of every calibration point receives $(-1)^k \zeta_R$, with $\zeta_R^2 = (r-1)/r$.
- *Pairs*: point j of every block of P_h receives $(-1)^j \zeta_h$, with $2r\zeta_h^2 = \lambda_{P_h\text{-within}}(\theta^\circ)$.
- *Squares*: point (a, a') receives $\zeta_a a + \zeta_{a'} a' + \zeta_{aa'} aa'$, with $4\zeta_a^2 = 4\zeta_{a'}^2 = \lambda_{\mathrm{Sq},a}(\theta^\circ)$ and $4\zeta_{aa'}^2 = \lambda_{\mathrm{Sq},aa'}(\theta^\circ)$.
- *Block means*: every observation of block b of family f receives $m_b = \pm\xi_f$, alternating in b, with $\frac{k_f r B}{B-1}\xi_f^2 = \lambda_{f\text{-between}}(\theta^\circ)$.
- *The complement W*: in the basis (gadget coordinates, u_f), let

$$y_W = \Sigma_W(\theta^\circ)\,\bar\alpha, \qquad \bar\alpha = (M\alpha^*,\ \beta_c,\ 0, \dots, 0),$$

  where Σ_W is Σ_θ restricted to W. The gadget observation at $(p, -1)$ is $(y_W)_p$, and every observation of family f receives the constant $c_f = (y_W)_f/\sqrt{n_f}$. The scalar β_c is chosen so that the data sum to exactly 0 (Lemma 12).

All magnitudes are rounded to rationals. The group magnitudes are rounded within a factor $e^{\pm\delta}$ (that is, a magnitude ζ is replaced by a rational in $[e^{-\delta}\zeta, e^{\delta}\zeta]$), using the same rational for both signs so every sign pattern stays exactly balanced, and the gadget values and the constants c_f ($f \neq P_1$) are rounded within ρ. The constant c_{P_1} is then taken to be the rational that makes the data sum to exactly 0. Since every other value moved by at most ρ, it differs from its unrounded value by at most $n\rho/n_{P_1} \le 5\rho$ (the ratio n/n_{P_1} is about 2 in the noisy model and 4 in the noise-free model). Let $\tilde y_W = \Sigma_W(\theta^\circ)\bar\alpha + e$ be the rounded projection, so $\|e_g\| \le \sqrt{n_0}\,\rho$ on the gadget coordinates and $\|e\| \le 6\sqrt{n}\,\rho$. Each group's empirical second moment s_g (the squared norm of the projection of y onto the group, divided by its dimension) then satisfies $s_g = \lambda_g(\theta^\circ)e^{\pm 2\delta}$, and the projection onto W is y_W up to rounding. At the target, before rounding, the posterior weights on W are exactly $\bar\alpha$: the gadget carries weight exactly $M\alpha^*$.

Lemma 12 (Offset)**.** *Suppose $n_{\max}\, g(4S/\Lambda) \le 1$. If $B \ge (2n_0+2)^2 M^2 \|\alpha^*\|_1^2$, then $|\beta_c| \le 1$.*

Proof. The group pieces and the replicate piece sum to zero. In the basis of W, a gadget coordinate contributes its value to the sum of the data, and u_f contributes $\sqrt{n_f}$ times its coefficient. The sum is then $\mathbf{1}_W^\top y_W$, where $\mathbf{1}_W = (\mathbf{1}_{n_0}, \sqrt{n_f})_f$ is the all-ones vector written in this basis, and it equals

$$\Sigma_{\mathrm{gad}} + c\,\beta_c, \quad c := \mathbf{1}_W^\top \Sigma_W(\theta^\circ) e_{u_1}, \quad \Sigma_{\mathrm{gad}} := \mathbf{1}_W^\top \Sigma_W(\theta^\circ)(M\alpha^*, 0).$$

Every entry of $\Sigma_W(\theta^\circ)$ in this basis is nonnegative and $u_1^\top \Sigma_{\theta^\circ} u_1 \ge 1$, so $c \ge \sqrt{n_1}$. Expand

$$\Sigma_{\mathrm{gad}} = \sum_{p,p'} A_{pp'} M\alpha^*_{p'} + \sum_f \sqrt{n_f} \sum_{p'} J_{p'f} M\alpha^*_{p'},$$

with A and J the gadget and coupling blocks of $\Sigma_W(\theta^\circ)$. Each column of A has absolute sum at most $n_0 + 1$. Each coupling entry satisfies $J_{p'f} \le \sqrt{n_f}\, g(4S/\Lambda)$, so

$$\sqrt{n_f}\, J_{p'f} \le n_{\max} g(4S/\Lambda) \le 1.$$

Note the factor $\sqrt{n_f}$ from $\mathbf{1}_W$, which is why the hypothesis is on $n_{\max}$ rather than $\sqrt{n_{\max}}$. Therefore

$$|\Sigma_{\mathrm{gad}}| \le (n_0 + 1 + |\mathcal{F}|)M\|\alpha^*\|_1 \le (2n_0 + 2)M\|\alpha^*\|_1,$$

and

$$|\beta_c| \le (2n_0 + 2)\frac{M\|\alpha^*\|_1}{\sqrt{n_1}} \le 1$$

since $n_1 \ge B$. □

Lemma 13 (Likelihood split)**.** *With n_g the dimension of group g,*

$$L(\theta) = -\tfrac{1}{2}\sum_g n_g[\log s_g + 1 + \psi(w_g)] + L_W(\theta), \quad w_g = \log\left(\frac{\lambda_g(\theta)}{s_g}\right), \quad \psi(w) = w + e^{-w} - 1,$$

where L_W is the Gaussian log-likelihood of the projection y_W under $\Sigma_W(\theta)$. Moreover,

- $\psi(w) \ge \frac{1}{6}\min(w^2, |w|)$ *for all w,*
- $\psi(w) \ge \frac{1}{4}|w|$ *for $|w| \ge 1$,*
- $\psi(w) \ge \frac{1}{4}(|w| - 1)$ *for all w, and*
- $\psi(w) \le w^2$ *for $|w| \le \frac{1}{2}$.*

Proof. Block-diagonalize Σ_θ with Lemma 11 and use

$$\log\lambda + s/\lambda = \log s + 1 + \psi(\log(\lambda/s))$$

on each group. For the bounds,

$$\psi(w) = w^2\int_0^1 (1-t)e^{-tw}\,\mathrm{d}t,$$

so $\psi(w)/w^2$ is decreasing on $\mathbb{R}$, and it equals $\frac{1}{2}$ at 0 and e^{-1} at $w = 1$. This gives

$$\psi(w) \ge \frac{w^2}{e} \ge \frac{w^2}{6} \quad \text{for} \quad w \le 1, \qquad \psi(w) \le w^2 \quad \text{for} \quad |w| \le \frac{1}{2}.$$

For $|w| \ge 1$, ψ is convex with $\psi' \ge 0.63$ on $[1, \infty)$, $\psi(1) = e^{-1}$, $\psi(-1) = e - 2$ and $\psi'(-1) = 1 - e$, which gives $\psi(w) \ge |w|/4$, implying the last bound. □

5 Abstract pinning

Let $\mathcal{W}_\eta$ be the set of θ with $|\ell^2/\Lambda - 1|,\ |\sigma_f^2 - 1|,\ |\sigma_n^2 - 1| \le \eta$, dropping the last coordinate in the noise-free model. Let

$$\varepsilon_{\mathrm{p}} = \frac{3m_2\eta}{100m_1\Lambda}.$$

Proposition 14 (Pinning)**.** *Suppose there are a function $\Psi(\theta) \ge 0$ and constants $\mathfrak{a}, \mathfrak{b}, \Gamma_0, \Psi_0 \ge 0$ with $\mathfrak{b} \ge 2\mathfrak{a}$ such that, for every θ:*

(H1) $L_W(\theta) - L_W(\theta^\circ) \le \Gamma_0 + \mathfrak{a}\,\Psi(\theta)$*;*

(H2) $\frac{1}{2}\sum_g n_g\psi(w_g(\theta)) \ge \mathfrak{b}\,(\Psi(\theta) - \Psi_0)$*;*

(H3) *if* $|\log(\lambda_g(\theta)/\lambda_g(\theta^\circ))| \le \varepsilon_{\mathrm{p}}$ *for every group, then* $\theta \in \mathcal{W}_\eta$.

Here $\mathfrak{b}$ *may grow with* B*, but* Ψ_0 *may not. Assume also* $\delta \le \min\{\varepsilon_{\mathrm{p}}/4, n^{-1/2}\}$*,* $n_g \ge B-1$ *for every group, and*

$$B - 1 \ge \frac{48\,\Gamma_1}{\varepsilon_{\mathrm{p}}^2}, \qquad \Gamma_1 := 4 + 2\Gamma_0 + 2\mathfrak{a}\Psi_0.$$

Then a joint MLE exists, and every joint MLE lies in $\mathcal{W}_\eta$.

Proof. Let $L(\theta) \ge L(\theta^\circ)$. Rounding gives $|w_g(\theta^\circ)| \le 2\delta \le \frac{1}{2}$, so

$$\frac{1}{2}\sum n_g \psi(w_g(\theta^\circ)) \le \frac{1}{2} n (2\delta)^2 \le 2.$$

Let $X := \frac{1}{2}\sum_g n_g \psi(w_g(\theta))$. Lemma 13 and (H1) give

$$X \le 2 + \Gamma_0 + \mathfrak{a}\Psi.$$

Combined with (H2),

$$(\mathfrak{b} - \mathfrak{a})\Psi \le 2 + \Gamma_0 + \mathfrak{b}\Psi_0,$$

and $\mathfrak{b} \ge 2\mathfrak{a}$ gives

$$\Psi \le 2 \cdot \frac{2 + \Gamma_0}{\mathfrak{b}} + 2\Psi_0.$$

Substituting back,

$$X \le 2(2 + \Gamma_0) + 2\mathfrak{a}\Psi_0 = \Gamma_1.$$

The slack $\mathfrak{b}\Psi_0$, which is proportional to B in the noise-free model, never enters Γ_1. Hence

$$\psi(w_g) \le \frac{2\Gamma_1}{B-1} \le \frac{(\varepsilon_{\mathrm{p}}/2)^2}{6}$$

for every group. By Lemma 13, this forces $|w_g| \le \varepsilon_{\mathrm{p}}/2$. Adding the rounding error $2\delta \le \varepsilon_{\mathrm{p}}/2$ gives $|\log(\lambda_g(\theta)/\lambda_g(\theta^\circ))| \le \varepsilon_{\mathrm{p}}$, and (H3) gives $\theta \in \mathcal{W}_\eta$. It follows that $\{L \ge L(\theta^\circ)\}$ lies in the compact set $\mathcal{W}_\eta$, where L is continuous. A maximizer exists, and every maximizer lies in $\mathcal{W}_\eta$. □

The proof shows more: every θ with $L(\theta) \ge L(\theta^\circ)$ lies in $\mathcal{W}_\eta$. In particular, a local maximizer reached by any ascent method started at θ° lies in $\mathcal{W}_\eta$. The same slack covers maximum a posteriori estimates.

Proposition 15 (MAP estimates)**.** *Let* Π *be a continuous hyperprior density on* θ *for the standardized data, with* $\sup\Pi < \infty$ *and* $\Pi(\theta_{\mathrm{s}}^\circ) > 0$*, and let*

$$\Gamma_\Pi := \log \frac{\sup \Pi}{\Pi(\theta_{\mathrm{s}}^\circ)}.$$

Suppose the hypotheses of Proposition 14 hold with Γ_1 *replaced by* $\Gamma_1 + 2\Gamma_\Pi$ *in the condition on* B*. Then a MAP estimate exists, and every MAP estimate lies in the window* $\mathcal{W}_\eta$ *rescaled by the standardizing factor.*

Proof. A prior breaks the scale equivariance of Lemma 4, so we work with the standardized data directly. By Lemma 4, their log-likelihood is L evaluated at the rescaled parameter, up to an additive constant, so Proposition 14 applies to it with $\theta_{\mathrm{s}}^\circ$ in place of θ°. If θ is a MAP estimate, or any θ with

$$L(\theta) + \log\Pi(\theta) \ge L(\theta_{\mathrm{s}}^\circ) + \log\Pi(\theta_{\mathrm{s}}^\circ),$$

then

$$L(\theta) - L(\theta_{\mathrm{s}}^{\circ}) \geq \log \frac{\Pi(\theta_{\mathrm{s}}^{\circ})}{\Pi(\theta)} \geq -\Gamma_{\Pi}.$$

The proof of Proposition 14 then runs with 2 replaced by $2 + \Gamma_{\Pi}$ in the bound on X, which gives $X \leq \Gamma_1 + 2\Gamma_{\Pi}$. The rest is unchanged. The set of such θ is therefore contained in the compact rescaled window, where $L + \log \Pi$ is continuous, so a maximizer exists. □

For log-normal hyperpriors, $\Gamma_{\Pi} = O((\log \Lambda)^2)$, and for hyperpriors with polynomial tails, $\Gamma_{\Pi} = O(\log \Lambda)$. The extra requirement on B is therefore only polylogarithmic in Λ. Every bound in Sections 7 and 8 holds at every θ in the window, so all results hold at MAP estimates as well.

Lemma 16 (Local slope of $\mathcal{R}$)**.** *There is $v_1 > 0$, depending only on $m_1, m_2, m_{2+\omega}$, such that $\mathcal{R}'(v) \leq -3m_2/m_1$ for $v \in (0, v_1]$.*

Proof. For $u \geq 0$ and $\omega \in (0, 1]$,

$$\left| e^{-u} - 1 + u - \frac{1}{2}u^2 \right| \leq \frac{1}{2}u^{2+\omega}, \quad |e^{-u} - 1 + u| \leq u^{1+\omega}.$$

Integrating against π gives

$$F(v) = m_1 v - \frac{1}{2}m_2 v^2 + O(v^{2+\omega}), \qquad F'(v) = m_1 - m_2 v + O(v^{1+\omega}),$$

with constants depending only on $m_{2+\omega}$. Substituting,

$$\mathcal{R}'(v) = \frac{4F'(4v)F(v) - F(4v)F'(v)}{F(v)^2} = \frac{-6m_1 m_2 v^2 + O(v^{2+\omega})}{m_1^2 v^2 (1 + O(v))} = -\frac{6m_2}{m_1} + O(v^{\omega}).$$ □

Lemma 17 (Identification)**.** *Let $v^{\circ} = 2/\Lambda$. Assume $2v^{\circ} \leq v_1$, $\Lambda \geq 16m_2/m_1$, and in the noisy model $r \geq \Lambda/m_1$. Then (H3) holds in both models.*

Proof. Let $v = 2/\ell^2$ and

$$\tilde{\kappa}_h = \sigma_f^2 \kappa_{P_h\text{-within}}(\ell) = \sigma_f^2 F(h^2 v) \quad \text{for} \quad h \in \{1, 2\},$$

so the P_h-within eigenvalue is $r\tilde{\kappa}_h + \sigma_n^2$. In the noisy model, the replicate group gives $\sigma_n^2 = e^{w'_R}$ with $|w'_R| \leq \varepsilon_{\mathrm{p}}$, and

$$r\,\tilde{\kappa}_h^{\circ} \geq rF(v^{\circ}) \geq rm_1 v^{\circ}/2 \geq 1$$

by Lemma 7(b). From $r\,\tilde{\kappa}_h + \sigma_n^2 = (r\,\tilde{\kappa}_h^{\circ} + 1)e^{w'_h}$ with $|w'_h| \leq \varepsilon_{\mathrm{p}}$,

$$\frac{\tilde{\kappa}_h}{\tilde{\kappa}_h^{\circ}} - 1 = (e^{w'_h} - 1) + \frac{1}{r\,\tilde{\kappa}_h^{\circ}}\left(e^{w'_h} - e^{w'_R}\right),$$

so $|\tilde{\kappa}_h/\tilde{\kappa}_h^{\circ} - 1| \leq 3.1\varepsilon_{\mathrm{p}}$; in the noise-free model the bound is $1.01\varepsilon_{\mathrm{p}}$. The ratio $\mathcal{R}(v) = \tilde{\kappa}_2/\tilde{\kappa}_1$ then satisfies $|\mathcal{R}(v) - \mathcal{R}(v^{\circ})| \leq 26\varepsilon_{\mathrm{p}}$, because $\mathcal{R} \leq 4$ by concavity of F. If $v \notin [v^{\circ}/2, 2v^{\circ}]$, then (A3) (monotonicity) and Lemma 16 on $(0, 2v^{\circ}] \subseteq (0, v_1]$ give

$$|\mathcal{R}(v) - \mathcal{R}(v^{\circ})| \geq \frac{3m_2}{m_1} \cdot \frac{v^{\circ}}{2} = \frac{3m_2}{m_1 \Lambda} > 26\varepsilon_{\mathrm{p}},$$

a contradiction. Inside the interval, the mean value theorem gives $|v - v^{\circ}| \leq 26\varepsilon_{\mathrm{p}} m_1/(3m_2)$, so

$$|v/v^{\circ} - 1| \leq \frac{13\varepsilon_{\mathrm{p}} m_1 \Lambda}{3m_2} = 0.13\eta, \quad |\ell^2/\Lambda - 1| \leq \eta/5.$$

Finally, $F' \le m_1$ and $F(v^\circ) \ge m_1 v^\circ/2$ give

$$|F(v)/F(v^\circ) - 1| \le 2|v/v^\circ - 1| \le 0.26\eta,$$

so $\sigma_f^2 = \tilde{\kappa}_1/F(v)$ satisfies

$$|\sigma_f^2 - 1| \le 3.1\varepsilon_{\mathrm{p}} + 0.27\eta \le \eta.$$

In the noisy model, $|\sigma_n^2 - 1| \le 2\varepsilon_{\mathrm{p}} \le \eta$. □

6 Checking (H1) and (H2)

In both models, $\mathfrak{a} = n_W/2$. The models differ in where the spectral floor of Σ_W comes from and in which groups are coercive. This section treats the noisy model; Appendix B treats the noise-free model.

	Noisy model	Noise-free model
Floor of $\Sigma_W(\theta)$	$\sigma_n^2 I$	$\sigma_f^2 \lambda_{\min}(K_W(\ell))$ (Lemma 45)
$\Psi(\theta)$	$\max(0, -w_R)$	$\log_+(1/\sigma_f^2) + 4\log_+(\ell^2/t_1)$
$\mathfrak{b}$	$n_R/8$	$B/32$
Ψ_0	1	$4c_4 + 20c_1 + 8$
Condition $\mathfrak{b} \ge 2\mathfrak{a}$	$n_R \ge 8n_W$	$B \ge 32n_W$

Common step for (H1). The quadratic form is nonnegative, and

$$\tilde{y}_W^\top \Sigma_W(\theta^\circ)^{-1} \tilde{y}_W = \bar{\alpha}^\top \Sigma_W(\theta^\circ)\bar{\alpha} + 2\bar{\alpha}^\top e + e^\top \Sigma_W(\theta^\circ)^{-1} e.$$

The last two terms are at most 0.01, and the final +1 below covers them. Indeed, $\bar{\alpha}^\top e = M\alpha^{*\top} e_g + \beta_c\, e_{u_1}$. With ρ as in Lemma 20, $\|e_g\| \le \sqrt{n_0}\,\rho$ and $\sqrt{n} \ge \sqrt{B} \ge (2n_0+2)M\|\alpha^*\|_1$ give

$$|M\alpha^{*\top} e_g| \le \frac{M\|\alpha^*\|_2\, \varepsilon_{\mathrm{w}} \lambda^-}{3\sqrt{n}} \le \frac{\varepsilon_{\mathrm{w}}}{3(2n_0+2)}.$$

The naïve bound $\|\bar{\alpha}\|\,\|e\|$ would not do here, because $\|\bar{\alpha}\|$ is of order $M\|\alpha^*\|_2$. Also,

$$|\beta_c e_{u_1}| \le \|e\| \le 6\sqrt{n}\,\rho \le 2\varepsilon_{\mathrm{w}}\lambda^-/n_0$$

by Lemma 12, and

$$e^\top \Sigma_W(\theta^\circ)^{-1} e \le 2\|e\|^2/\lambda^-$$

because $\lambda_{\min}(\Sigma_W(\theta^\circ)) \ge \lambda^-/2$. With $\varepsilon_{\mathrm{w}} = 0.05$, $\lambda^- \le 1$ and $n_0 \ge 25$, the total is at most 0.01. Hence

$$L_W(\theta) - L_W(\theta^\circ) \le \frac{1}{2} n_W [\log \lambda_{\max}(\Sigma_W(\theta^\circ)) - \log \lambda_{\min}(\Sigma_W(\theta))] + \frac{1}{2}\lambda_{\max}(\Sigma_W(\theta^\circ))\|\bar{\alpha}\|^2 + 1. \quad (2)$$

Assume $n_{\max}\, g(4S/\Lambda) \le 1$. By Gershgorin, a gadget row of $\Sigma_W(\theta^\circ)$ has absolute row sum at most $n_0 + 1 + |\mathcal{F}|$, and a family row at most

$$k_f r + 1 + n_{\max} g(8S/\Lambda) + n_0 \le 4r + 2 + n_0.$$

Therefore,

$$\lambda_{\max}(\Sigma_W(\theta^\circ)) \le \bar{\lambda}_W := n_0 + 4r + 4,$$

with $r = 1$ in the noise-free model.

Lemma 18 (Noisy model)**.** *(H1) and (H2) hold with*

$$\Psi = \max\{0, -w_R\}, \quad \Gamma_0 = \frac{1}{2} n_W (\log \bar{\lambda}_W + 1) + \frac{1}{2} \bar{\lambda}_W \|\bar{\alpha}\|^2 + 1, \quad \mathfrak{b} = \frac{n_R}{8}, \quad \Psi_0 = 1.$$

Proof. For (H1), $\lambda_{\min}(\Sigma_W(\theta)) \geq \sigma_n^2$, and

$$\log(1/\sigma_n^2) \leq -w_R + 2\delta \leq \Psi + \frac{1}{2}$$

because $s_R \geq e^{-2\delta}$ by rounding. For (H2), keep only the replicate group. By Lemma 13,

$$\frac{1}{2} n_R \psi(w_R) \geq \frac{n_R}{8}(|w_R| - 1) \geq \frac{n_R}{8}(\Psi - 1). \qquad \square$$

In the noise-free model, Lemma 45 supplies the spectral floor and Lemma 47 checks (H1) and (H2); both are in Appendix B.

7 The posterior mean in the window

Fix $\theta \in \mathcal{W}_\eta$. By Lemma 11,

$$\Sigma_\theta^{-1} y = \sum_g \frac{y_g}{\lambda_g(\theta)} + \alpha_W(\theta),$$

where y_g is the projection of y onto group g and $\alpha_W = \Sigma_W(\theta)^{-1} \tilde{y}_W$, with $\tilde{y}_W$ the rounded y_W. Accordingly,

$$\begin{aligned}
\mu_\theta &= \mu_{\text{gad}} + \mu_u + \mu_{\text{grp}}, \\
\mu_{\text{gad}}(x) &= \sigma_f^2 \sum_p (\alpha_W)_p k_\ell(x, (p, -1)), \\
\mu_u(x) &= \sigma_f^2 \sum_f (\alpha_W)_f \, k_\ell(x)^\top u_f, \\
\mu_{\text{grp}}(x) &= \sigma_f^2 \sum_g \frac{k_\ell(x)^\top y_g}{\lambda_g(\theta)}.
\end{aligned}$$

Replicate contrasts contribute nothing, because $k_\ell(x)$ is constant across replicates. Now let

$$\Sigma_W(\theta) = \begin{bmatrix} A_\theta & J_\theta \\ J_\theta^\top & U_\theta \end{bmatrix},$$

with A_θ the $n_0 \times n_0$ gadget block, U_θ the $|\mathcal{F}| \times |\mathcal{F}|$ family-sum block and J_θ the coupling. Define

$$\lambda^- := \begin{cases} 0.98 & \text{noisy model}, \\ 0.98\, \lambda_G^- & \text{noise-free model}, \end{cases}$$

with λ_G^- as in the proof of Theorem 2, and $\epsilon_C := 1.02\sqrt{3n_0 n}\; g(2S/\Lambda)$.

Lemma 19 (Blocks)**.** *Assume S satisfies (3) below. For every $\theta \in \mathcal{W}_\eta$, we have the following:*

$$\lambda_{\min}(A_\theta) \geq \lambda^-, \quad \|A_\theta - A_{\theta^\circ}\| \leq 1.6 n_0 \eta, \quad \|J_\theta\| \leq \epsilon_C, \quad \lambda_{\min}(U_\theta) \geq 0.9r, \quad \|U_\theta - U_{\theta^\circ}\| \leq 9r\eta.$$

Proof. In the noisy model, $A_\theta \succeq \sigma_n^2 I \succeq 0.98I$. In the noise-free model, $A_\theta = \sigma_f^2 K_{GG}(\ell)$, and Corollary 50 with $\ell^2 \le 2\Lambda$ gives $\lambda_{\min}(K_{GG}(\ell)) \ge \lambda_G^-$. Next,

$$A_\theta - A_{\theta^\circ} = \sigma_f^2(K_{GG}(\ell) - K_{GG}(\Lambda)) + (\sigma_f^2 - 1)K_{GG}(\Lambda) + (\sigma_n^2 - 1)I,$$

and Lemma 7(e) bounds the first term by $1.02 \cdot 0.38 n_0 \eta$ and Gershgorin bounds the second by $n_0\eta$, so the total is at most $1.6 n_0 \eta$. An entry of J_θ is

$$\frac{\sigma_f^2}{\sqrt{n_f}} \sum_{x \in f} k_\ell((p,-1),x) \le 1.02\sqrt{n_f}\, g(4S/\ell^2),$$

and $\ell^2 \le 2\Lambda$; the Frobenius norm gives $\|J_\theta\| \le \epsilon_C$. A diagonal entry of U_θ is

$$u_f^\top \Sigma_\theta u_f = \sigma_f^2 n_f^{-1} \mathbf{1}_f^\top K_{ff} \mathbf{1}_f + \sigma_n^2.$$

All entries of K_{ff} are nonnegative, and the $(k_f r)^2$ entries inside each block are at least $g(8/\ell^2) \ge 0.999$, so the diagonal entry is at least $0.98 \cdot 0.999\, k_f r \ge 1.9r$. An off-diagonal entry is at most

$$1.02\sqrt{n_f n_{f'}}\, g(8S/\ell^2) \le 1.02\, n\, g(S/(2\Lambda)) \le 0.001$$

by (3), so Gershgorin gives $\lambda_{\min}(U_\theta) \ge 1.9r - 0.002 \ge 0.9r$. For the change across the window, split $n_f^{-1}\mathbf{1}_f^\top K_{ff}\mathbf{1}_f$ into the within-block part, at most $k_f r$, and the between-block part, at most $n_f g(8S/\ell^2) \le 0.001$. Then

$$\begin{aligned} |(U_\theta - U_{\theta^\circ})_{ff}| &\le \overbrace{\eta(k_f r + 0.001)}^{\sigma_f^2 \text{ change}} + \overbrace{1.02 \cdot 0.38\, \eta\, k_f r}^{\text{(e), first bound}} + \overbrace{1.02 \cdot 0.76\, \eta\, n_f g(8S/(3\Lambda))}^{\text{(e), second bound}} + \overbrace{\eta}^{\sigma_n^2 \text{ change}} \\ &\le 1.4\, k_f r\, \eta + 1.01\, \eta, \end{aligned}$$

because $n_f g(8S/(3\Lambda)) \le n\, g(S/(2\Lambda)) \le 0.001$. The first bound alone would not do here: it would charge 0.38η to each of the roughly n_f^2 between-block pairs, giving $0.38\eta\, n_f$. With $k_f r \le 4r$ the diagonal change is at most $6.6r\eta$, and by the second bound, the off-diagonal changes are at most 0.001η each, so $\|U_\theta - U_{\theta^\circ}\| \le 9r\eta$. □

Lemma 20 (Weights)**.** *Let*

$$Y := \bar{\lambda}_W(M\|\alpha^*\|_2 + 1) + 1, \quad A_{\max} := \frac{2Y}{\lambda^-},$$

and let $\varepsilon_{\mathrm{w}} \in (0,1]$. *Choose*

$$\eta \le \min\left\{0.02, \frac{\varepsilon_{\mathrm{w}}\lambda^-}{5n_0^{3/2}M\|\alpha^*\|_2}\right\}, \quad \rho \le \frac{\varepsilon_{\mathrm{w}}\lambda^-}{3n_0\sqrt{n}}, \quad \epsilon_C\,(M\|\alpha^*\|_2 + A_{\max} + 3) \le \frac{\varepsilon_{\mathrm{w}}\lambda^-}{9\sqrt{n_0}}.$$

Then for every $\theta \in \mathcal{W}_\eta$,

$$\|(\alpha_W)_{\mathrm{gad}} - M\alpha^*\|_1 \le \varepsilon_{\mathrm{w}}$$

and every family-sum weight satisfies $|(\alpha_W)_f| \le 2$.

Proof. Let $\alpha_W = (\alpha_g, \alpha_u)$ and $\tilde{y}_W = \Sigma_W(\theta^\circ)\bar{\alpha} + e$, with $\|e_g\| \le \sqrt{n_0}\,\rho$ and $\|e\| \le 6\sqrt{n}\,\rho$ (see the rounding paragraph in Section 4). The two block equations of $\Sigma_W(\theta)\alpha_W = \tilde{y}_W$ read

$$A_\theta \alpha_g + J_\theta \alpha_u = A_{\theta^\circ} M\alpha^* + J_{\theta^\circ}\beta_c e_1 + e_g,$$
$$J_\theta^\top \alpha_g + U_\theta \alpha_u = J_{\theta^\circ}^\top M\alpha^* + U_{\theta^\circ}\beta_c e_1 + e_u.$$

A priori bound. Viewing $\Sigma_W(\theta)$ as its block diagonal plus the off-diagonal blocks, which have norm $\|J_\theta\|$, Weyl's inequality and Lemma 19 give

$$\lambda_{\min}(\Sigma_W(\theta)) \ge \min\{\lambda^-, 0.9r\} - \epsilon_C = \lambda^- - \epsilon_C \ge \frac{\lambda^-}{2},$$

since $\lambda^- \le 0.98 < 0.9r$ in the noisy model ($r \ge 10^4$) and $\lambda^- < 0.9$ in the noise-free model. Since $\|\tilde{y}_W\| \le \bar{\lambda}_W\|\bar{\alpha}\| + 1 \le Y$, we get $\|\alpha_W\| \le A_{\max}$.

Family-sum weights. From the second equation,

$$\alpha_u - \beta_c e_1 = U_\theta^{-1}\big[(U_{\theta^\circ} - U_\theta)\beta_c e_1 + J_{\theta^\circ}^\top M\alpha^* - J_\theta^\top \alpha_g + e_u\big],$$

so by Lemma 19, Lemma 12 and the a priori bound,

$$\|\alpha_u - \beta_c e_1\| \le \frac{1}{0.9r}\big(9r\eta + \epsilon_C(M\|\alpha^*\|_2 + A_{\max}) + 6\sqrt{n}\rho\big) \le 10\eta + 0.01 \le 0.5.$$

Hence $|(\alpha_W)_f| \le |\beta_c| + 0.5 \le 2$.

Gadget weights. From the first equation,

$$\alpha_g - M\alpha^* = A_\theta^{-1}\big[(A_{\theta^\circ} - A_\theta)M\alpha^* + J_{\theta^\circ}\beta_c e_1 - J_\theta\alpha_u + e_g\big],$$

so

$$\|\alpha_g - M\alpha^*\|_2 \le \frac{1}{\lambda^-}(1.6n_0\eta M\|\alpha^*\|_2 + 3\epsilon_C + \sqrt{n_0}\,\rho) \le \frac{\varepsilon_{\mathrm{w}}}{3\sqrt{n_0}} + \frac{\varepsilon_{\mathrm{w}}}{3\sqrt{n_0}} + \frac{\varepsilon_{\mathrm{w}}}{3\sqrt{n_0}}.$$

The ℓ^1 bound follows from $\|\cdot\|_1 \le \sqrt{n_0}\|\cdot\|_2$. □

Lemma 21 (Contributions)**.** *Use $\varepsilon_{\mathrm{m}} = 0.01$ in Proposition 8 and $\varepsilon_{\mathrm{w}} = 0.05$ in Lemma 20. Assume*

$$\Lambda \ge \max\{12800m_1 + 10^4\sqrt{m_2},\ 16m_2/m_1\},$$

$r \ge 10^4$ in the noisy model, $B \ge 10^6 r$, and S so large that

$$n^2(1 + M\|\alpha^*\|_1)(1 + \Lambda/m_1)\,g\big(S/(2\Lambda)\big) \le 0.001. \tag{3}$$

Let $\Omega_0 = \{(z, -1)\}$ be the gadget face.

(a) $\mu_{\mathrm{grp}} = 0$ *on* Ω_0.

(b) $|\mu_u| \le 0.02$ *on* $[-1, 1]^D$.

(c) *In yes instances,* $\mu_{\mathrm{gad}}(v^*, -1) \le -90.3$. *In no instances,* $\mu_{\mathrm{gad}} \ge -0.1$ *on* $[-1, 1]^D$.

(d) $|\mu_{\mathrm{grp}}| \le 1.1$ *on* $[-1, 1]^D$.

Proof. We will prove each component one at a time.

(a) On Ω_0, x is equidistant from all points of each block and of each family, and each y_g sums to zero on each block or family.

(b) Let $D_b(x)$ be the squared distance from x to the base of block b, the point with main coordinates 0, hub coordinates $+1$ on S_b, and -1 elsewhere. Bases of different blocks are at squared distance $8S$, so at most one block has $D_b(x) < 2S$. A point p of block b differs from the base of b in at most two private coordinates of b, each by $h \le 2$, and every coordinate of x lies in $[-1, 1]$, so

$$\big|\|x - p\|^2 - D_b(x)\big| \le 16.$$

Since $S \ge 16$, every point of every other block is at squared distance at least $2S - 16 \ge S$ from x. Hence

$$|k_\ell(x)^\top u_f| \le n_f^{-1/2}\big(k_f r + n_f g(S/\ell^2)\big) \le \sqrt{4r/B} + \sqrt{n}\, g(S/(2\Lambda)).$$

With $|(\alpha_W)_f| \le 2$ (Lemma 20), $|\mathcal{F}| \le 3$ and $\sigma_f^2 \le 1.02$, $|\mu_u| \le 6.12 \cdot 0.003 < 0.02$.

(c) All gadget points share their hub and private coordinates, so at $x = (z, z') \in [-1, 1]^D$, with z' the hub and private coordinates, every squared distance to a gadget point exceeds its value on the face by the same $t = \|z' + 1\|^2$. Hence

$$\sum_p (M\alpha^*)_p k_\ell(x, (p, -1)) = M \int e^{-\tau t} G_\tau(z)\, \mathrm{d}\pi_\ell(\tau).$$

Apply Proposition 8 (valid since $|\ell^2/\Lambda - 1| \le \eta \le 0.02$), Lemma 20 with $k_\ell \le 1$, and $\sigma_f^2 \in [0.98, 1.02]$, bounding each sign on its own side. With $M\gamma = 4$,

$$\text{yes:}\ \ -0.98 \cdot 23.05 \cdot 4 + 1.02 \cdot 0.05 \le -90.3, \qquad \text{no:}\ \ -1.02\,(0.011 \cdot 4 + 0.05) \ge -0.1.$$

(d) Appendix C. □

Proof of Theorem 3

The parameters are chosen in the order of the table below, subject to these constraints.

1. $\Lambda \ge 2T_\star/\tau_0$ (Proposition 8 with $\varepsilon_{\mathrm{m}} = 0.01$), $2v^\circ \le v_1$ (Lemma 17), $v^\circ \le m_1/(18m_2)$ and $\Lambda \ge 4T_1$ (Lemma 47), and the conditions of Lemma 21.
2. η as in Lemma 20 with $\varepsilon_{\mathrm{w}} = 0.05$.
3. r even, $r \ge \max\{\Lambda/m_1, 10^4\}$ (noisy model).
4. B even, $B \ge 10^6 r$, $B \ge (2n_0 + 2)^2 M^2 \|\alpha^*\|_1^2$, $B \ge 32 n_W$ (noise-free model; $n_R \ge 8n_W$ is automatic in the noisy model), $B \ge 10^4 n_0 (n_0 + 2)^2 (1 + M\|\alpha^*\|_1)^2$, and $B - 1 \ge 48\Gamma_1/\varepsilon_{\mathrm{p}}^2$.
5. $S \ge 16$, $n_{\max} g(4S/\Lambda) \le 1$, the bound on ϵ_C in Lemma 20, and (3).
6. $\delta \le \min\{\varepsilon_{\mathrm{p}}/4, n^{-1/2}\}$, and ρ as in Lemma 20.

Each condition on S after the first has the form $N\, g(cS/\Lambda) \le c'$ with N polynomial and c, c' absolute constants. By (A2) it holds once

$$S \ge \frac{\Lambda}{c}\left(\frac{C_g N}{c'}\right)^{1/\vartheta}.$$

These values are polynomial.

The choice of B and S in the noise-free model. Here item 4 depends on item 5, because Γ_1 grows with S through P_W. Write $S(B)$ for the smallest integer satisfying item 5 for a given B; since $n = n_0 + 8B$, every N in item 5 is at most $C_N d^{k_N} B^{k_N}$ for constants C_N, k_N. By Lemma 47, $\Gamma_1 = \Gamma_1' + n_W P_W$, where Γ_1' does not depend on B or S, and by Lemma 45, $P_W \le C_W S \log n$ in general and $P_W \le C_W \log(nS)$ under (A5)(ii).

- Under (A5)(i), $S(B) \le C\Lambda\big(\log(dB)\big)^{1/\vartheta'}$, so $P_W \le C'\Lambda\big(\log(dB)\big)^{1+1/\vartheta'}$.
- Under (A5)(ii), $S(B)$ is polynomial in dB by (A2), so $P_W \le C' \log(dB)$.

In both cases the last condition of item 4 reads $B - 1 \ge a + b\,(\log(dB))^k$ with a, b polynomial in d and k a constant depending only on g. Since the right side grows polylogarithmically in B, it holds for every $B \ge B_0$ with B_0 polynomial in d. We take B to be the smallest even integer that is at least B_0 and meets the other conditions of item 4, and then $S = S(B)$. Remark 46 explains why this needs (A5).

Symbol	Role	Constraint
Λ	target ℓ^2; makes $\tau \le \tau_0$ carry almost all of π_Λ	1
$\gamma,\ M$	$\gamma = m_2/\Lambda^2$, gadget amplitude $M = 4/\gamma$	set by Λ
$t_1,\ p_1,\ \lambda^-$	noise-free spectral floor	set by Λ
$\eta,\ \varepsilon_{\mathrm{p}}$	width of the window $\mathcal{W}_\eta$; pinning tolerance	2
r	replicates per calibration point (noisy model)	3
B	blocks per family; pinning strength	4
S	hub size; separates blocks from each other and from the gadget	5
$\delta,\ \rho$	rounding of group magnitudes; of gadget values and c_f	6
$\varepsilon_{\mathrm{m}},\ \varepsilon_{\mathrm{w}}$	slack in Proposition 8; gadget weight error	fixed: 0.01, 0.05

All quantities are polynomial in d, with constants depending only on g, so the instance (with $n = n_0 + r\sum_f k_f B$ points in dimension $D \le d + |\mathcal{F}|B(S+4)$) has polynomial size and bit length, and (A4) makes it computable in polynomial time. By Proposition 14 with Lemmas 17, 18 and 47, a joint MLE exists and every joint MLE lies in $\mathcal{W}_\eta$. By Lemma 21:

- in yes instances, $\min \mu \le \mu(v^*, -1) \le -90.3 + 0.02 \le -90.28$;
- in no instances, $\min \mu \ge -0.1 - 1.1 - 0.02 \ge -1.22$.

Strong sense. Every number in the instance is a rational with numerator and denominator bounded by a polynomial in d. The coordinates lie in $\{-1, 0, 1\}$ for calibration points and in $\{-1, 0, \pm\frac{1}{2}, 1\}$ for gadget points. The group magnitudes are rounded within a factor $e^{\pm\delta}$, and the gadget values and the constants c_f are multiples of ρ, all of polynomial size. The standardizing factor below is a rational with polynomial denominator. Kernel values are computed to higher precision inside the reduction, but they do not appear in the instance.

Standardization. The data sum to exactly 0 by the choice of β_c. Each calibration observation averages $c_f^2 + \xi_f^2$ plus the squared contrast magnitudes over its sign pattern because all cross terms cancel. Here, $c_f = (y_W)_f/\sqrt{n_f}$ is $O(\sqrt{r/B})$, so $c_f^2 \le 10^{-5}$. In the noisy model,

$$\xi_f^2 \in [0.999,\, 1.0001], \qquad \zeta_h^2 \le 0.001, \qquad \zeta_R^2 \in [0.9999,\, 1],$$

because $\lambda_{P_h\text{-between}}(\theta^\circ) \in [1.999r + 1,\, 2r + 1]$ and $F(2h^2/\Lambda) \le 8m_1/\Lambda \le 0.001$. In the noise-free model,

$$\xi_f^2 \in [0.999,\, 1], \qquad \text{every contrast magnitude squared is at most } 0.001,$$

because $\kappa_{f\text{-between}}(\Lambda)/k_f \in [0.9995,\, 1]$. Each gadget value satisfies

$$|(y_W)_p| \le (n_0 + 2)(1 + M\|\alpha^*\|_1),$$

so the gadget share of the mean square is at most

$$n_0(n_0 + 2)^2(1 + M\|\alpha^*\|_1)^2/n \le 0.001.$$

Hence the mean square of y lies in $[1.998, 2.003]$ in the noisy model and in $[0.998, 1.004]$ in the noise-free model, so

$$\mathrm{sd}(y) \in [1.41, 1.42] \text{ (noisy)}, \qquad \mathrm{sd}(y) \in [0.99, 1.01] \text{ (noise-free)}.$$

Let c_{s} be a rational with $\mathrm{sd}(y) \le c_{\mathrm{s}} \le \mathrm{sd}(y)(1 + \frac{1}{4n})$. Dividing by c_{s} gives $\frac{1}{n}\sum_i y_i^2 \in [1 - \frac{1}{n}, 1]$, and c_{s} lies in the same ranges. Applying Lemma 4:

- yes: $\min \mu \le -90.28/1.42 \le -63$ (noisy) and $\le -90.28/1.01 \le -89$ (noise-free);
- no: $\min \mu \ge -1.22/1.41 \ge -0.87$ (noisy) and $\ge -1.22/0.99 \ge -1.25$ (noise-free).

In both models, therefore, $\min \mu \le -63$ in yes instances and $\min \mu \ge -1.25$ in no instances. □

8 Acquisition functions and ε-optimality

Throughout, μ and σ are the posterior mean and standard deviation of the latent function at a joint MLE, in standardized units. Recall the levels $c_{\mathrm{yes}} = 63$ and $c_{\mathrm{no}} = 1.25$ of Theorem 3. In yes instances $\min \mu \le -c_{\mathrm{yes}}$, and in no instances $\min \mu \ge -c_{\mathrm{no}}$, at every $\theta \in \mathcal{W}_\eta$.

Lemma 22 (Posterior standard deviation)**.** $0 \le \sigma(x) \le \bar{s} := 1.03$ *for all* $x \in [-1, 1]^D$.

Proof. Posterior variance is at most prior variance, $\hat{\sigma}_f^2 \le 1 + \eta \le 1.02$ before standardization, and standardization divides $\hat{\sigma}_f^2$ by $c_{\mathrm{s}}^2 \ge 0.99^2$ (Lemma 4); so $\hat{\sigma}_f \le 1.03$. □

In the noisy model $c_{\mathrm{s}} \ge 1.41$, so in fact $\hat{\sigma}_f \le 0.72$. If an acquisition function uses the predictive standard deviation of a noisy observation, $\sqrt{\sigma^2 + \hat{\sigma}_n^2}$, the same argument gives at most

$$\frac{\sqrt{2.04}}{1.41} \le 1.02,$$

so the bounds below apply unchanged.

Lemma 23 (Transfer)**.** *Let $a : \mathbb{R} \times [0, \bar{s}] \to \mathbb{R}$ be nondecreasing in its first argument, and let*

$$a_- = \sup_{s\in[0,\bar{s}]} a(-c_{\text{yes}}, s), \qquad a_+ = \inf_{s\in[0,\bar{s}]} a(-c_{\text{no}}, s).$$

If $a_- < a_+$, the reduction maps yes instances to instances with $\min A \le a_-$ and no instances to instances with $\min A \ge a_+$, for $A(x) = a(\mu(x), \sigma(x))$ over $[-1, 1]^D$. Deciding between the two is therefore NP-hard.

Proof. In yes instances some x has $\mu(x) \le -c_{\text{yes}}$, so by monotonicity and Lemma 22, $A(x) \le a(-c_{\text{yes}}, \sigma(x)) \le a_-$. In no instances every x has $\mu(x) \ge -c_{\text{no}}$, so $A(x) \ge a(-c_{\text{no}}, \sigma(x)) \ge a_+$. □

The lemma needs no information about σ at the witness; it uses only that σ takes values in $[0, \bar{s}]$. It also needs nothing about the particular values c_{yes} and c_{no}, so any improvement in Theorem 3 carries over directly.

Lemma 24 (Two-sided EI bound)**.** *Let $\tilde{\Phi}(z) = z\Phi(z) + \phi(z)$, where Φ and ϕ are the standard normal cdf and pdf. Then $\tilde{\Phi}' = \Phi$, so $\tilde{\Phi}$ is strictly increasing, $\tilde{\Phi}(z) \to 0^+$ as $z \to -\infty$, and $\tilde{\Phi}(z) - z \to 0$ as $z \to \infty$. For every $z \in \mathbb{R}$, $\max\{z, 0\} \le \tilde{\Phi}(z) \le \max\{z, 0\} + \phi(0)$.*

Proof. For $z > 0$, $\tilde{\Phi}(z) - z = \phi(z) - z\Phi(-z)$. The Mills-ratio bound $\Phi(-z) \le \phi(z)/z$ gives $z\Phi(-z) \le \phi(z)$, so $\tilde{\Phi}(z) - z \ge 0$; and $z\Phi(-z) \ge 0$ gives $\tilde{\Phi}(z) - z \le \phi(z) \le \phi(0)$. For $z \le 0$, $\tilde{\Phi}(z) > 0$ because $\tilde{\Phi}$ is increasing with limit 0 at $-\infty$, and $\tilde{\Phi}(z) = z\Phi(z) + \phi(z) \le \phi(z) \le \phi(0)$. □

For minimization with incumbent f, expected improvement is $\text{EI}(x) = \sigma(x)\tilde{\Phi}(z_f(x))$ with

$$z_f(x) = \frac{f - \mu(x)}{\sigma(x)},$$

extended to $\sigma(x) = 0$ by $\text{EI} = \max\{f - \mu, 0\}$. Lemma 24 gives

$$\max\{f - \mu(x), 0\} \le \text{EI}(x) \le \max\{f - \mu(x), 0\} + \sigma(x)\phi(0). \tag{4}$$

Probability of improvement is $\text{PI}(x) = \Phi(z_f(x))$, extended by $\mathbf{1}[\mu(x) < f]$ at $\sigma(x) = 0$.

Gittins index. For minimization with cost $c_0 > 0$, the index $I_G(x)$ solves $\sigma\tilde{\Phi}((I_G - \mu)/\sigma) = c_0$, so

$$I_G(x) = \mu(x) + h(\sigma(x)), \qquad h(s) = s\,\tilde{\Phi}^{-1}(c_0/s).$$

$\tilde{\Phi} : \mathbb{R} \to (0, \infty)$ is a strictly increasing bijection, so $\tilde{\Phi}^{-1}$ is well defined. As $s \to 0^+$, $c_0/s \to \infty$ and $\tilde{\Phi}^{-1}(w) - w \to 0$, so $h(s) = s(c_0/s + o(1)) \to c_0$. Hence h extends continuously to $[0, \bar{s}]$ with $h(0) = c_0$. With $z = \tilde{\Phi}^{-1}(c_0/s)$, the derivative is $h'(s) = -\phi(z)/\Phi(z) < 0$, so h is decreasing, and its oscillation over $[0, \bar{s}]$ is

$$\text{osc}(c_0) := \sup_{[0,\bar{s}]} h - \inf_{[0,\bar{s}]} h = c_0 - h(\bar{s}).$$

What matters for the transfer below is this oscillation, not $\sup|h|$: adding a constant to h shifts both levels of Lemma 23 equally.

Lemma 25 (Gittins oscillation)**.** $\text{osc}(c_0)$ *is strictly decreasing in c_0, tends to 0 as $c_0 \to \infty$, and for every $c_0 > 0$,*

$$\text{osc}(c_0) \le 0.42 + 1.03\sqrt{2\log_+(0.42/c_0)}.$$

In particular $\text{osc}(c_0) < 61.74$ for $c_0 \ge 10^{-770}$, $\text{osc}(c_0) < 61.35$ for $c_0 \ge 10^{-760}$, and $\text{osc}(c_0) \le \sqrt{\text{poly}}$ whenever c_0 is a rational of polynomial bit length.

Proof. Let $z_* = \tilde{\Phi}^{-1}(c_0/\bar{s})$, so $h(\bar{s}) = \bar{s}z_*$. Differentiating $\tilde{\Phi}(z_*) = c_0/\bar{s}$ gives

$$\mathrm{d}(\bar{s}z_*)/\mathrm{d}c_0 = 1/\Phi(z_*) > 1,$$

so

$$\mathrm{d}\,\mathrm{osc}/\mathrm{d}c_0 = 1 - 1/\Phi(z_*) < 0.$$

As $c_0 \to \infty$, $z_* - c_0/\bar{s} \to 0$ by Lemma 24, so osc $\to 0$. If $c_0 \geq \bar{s}\phi(0)$ (note $\bar{s}\phi(0) \leq 0.411$), then by monotonicity

$$\mathrm{osc}(c_0) \leq \mathrm{osc}(\bar{s}\phi(0)) = \bar{s}\phi(0) \leq 0.42,$$

because $\tilde{\Phi}(0) = \phi(0)$ gives $h(\bar{s}) = 0$ at $c_0 = \bar{s}\phi(0)$. If $c_0 < \bar{s}\phi(0)$, then $z_* < 0$, and $\tilde{\Phi}(z) \leq \phi(z)$ for $z \leq 0$ (Lemma 24) gives $\phi(z_*) \geq c_0/\bar{s}$; that is,

$$|z_*| \leq \sqrt{2\log(\bar{s}\phi(0)/c_0)} \leq \sqrt{2\log(0.42/c_0)}.$$

Hence

$$\mathrm{osc}(c_0) \leq c_0 + \bar{s}|z_*| \leq 0.42 + 1.03\sqrt{2\log(0.42/c_0)}.$$

The numerical consequences follow by evaluating the bound at 10^{-770} and 10^{-760}, and the last claim from $\log(1/c_0) \leq$ poly. □

For comparison, a direct evaluation gives $\mathrm{osc}(10^{-180}) \approx 29.4$, $\mathrm{osc}(1) \approx 0.11$ and $\mathrm{osc}(5) < 10^{-6}$.

Corollary 26. *Each problem below is NP-hard. The incumbent f is part of the input, except in the row marked* $\min_i y_i$.

Acquisition	*Objective*	*Covered when*
Additive σ-term	min $\mu - h(\sigma)$	$\sup h - \inf h < 61.75$
LCB	min $\mu - \beta^{1/2}\sigma$	$\beta^{1/2} \leq 59$
Gittins index	min I_G	$\mathrm{osc}(c_0) < 61.75$*; e.g. every* $c_0 \geq 10^{-770}$
EI, incumbent f	max EI	$f > -62.58$
EI, incumbent $\min_i y_i$	max EI	*noise-free model (Proposition 28)*
PI, incumbent f	max PI	$f \in (-63, -1.25)$

Proof. We will prove the result for each acquisition function separately.

- Additive term: Apply Lemma 23 with $a(m, s) = m - h(s)$, so

  $$a_- = -63 - \inf h < -1.25 - \sup h = a_+$$

  exactly when $\sup h - \inf h < 61.75$. In particular, $|h| \leq H$ with $H < 30.875$ suffices.

- LCB: $a_- = -63$ (at $s = 0$) and $a_+ = -1.25 - 1.03\,\beta^{1/2} \geq -62.02 > a_-$.

- Gittins: The additive case with $-h$ in place of h, so $a_- = -63 + c_0$ and $a_+ = -1.25 + h(\bar{s})$, which are separated exactly when $\mathrm{osc}(c_0) < 61.75$; Lemma 25 gives this for every $c_0 \geq 10^{-770}$.

- EI: By (4), in yes instances, some x^* has $\mathrm{EI}(x^*) \geq f + 63 > 0$, and in no instances,

  $$\mathrm{EI}(x) \leq \max\{f + 1.25, 0\} + 1.03\phi(0) \leq \max\{f + 1.25, 0\} + 0.42 \quad \forall\, x.$$

  These are separated: if $f \geq -1.25$, it reads $f + 63 > f + 1.67$, and if $f < -1.25$, it reads $f + 63 > 0.42$, that is, $f > -62.58$.

- EI with $f = \min_i y_i$: in the noise-free model this f lies in $[-1.04, -0.96]$ by Proposition 28, so the previous case applies.
- PI: If $\mu(x^*) \le -63 < f$ then $z_f(x^*) \ge (f+63)/\bar{s}$, and $\mathrm{PI}(x^*) = 1$ if $\sigma(x^*) = 0$. If $\mu(x) \ge -1.25 > f$ for every x, then $z_f(x) \le (f+1.25)/\bar{s}$, and $\mathrm{PI}(x) = 0$ if $\sigma(x) = 0$. Since $f + 63 > 0 > f + 1.25$, the thresholds $\Phi((f+63)/\bar{s})$ and $\Phi((f+1.25)/\bar{s})$ are separated; for $f = -32$ they are $1-\epsilon$ and ϵ with $\epsilon < 10^{-100}$. □

By Lemma 4, rescaling y rescales μ, σ and $\hat{\sigma}_f$ together. The covered ranges are therefore statements about $f/\hat{\sigma}_f$, β and $c_0/\hat{\sigma}_f$, and no rescaling of the data moves an incumbent into or out of them.

The incumbent. The two models differ here.

- In the noisy model, the gadget observations are $(y_W)_p = \mu(p) + \sigma_n^2 M \alpha_p^*$ at the target. Several weights α_p^* are negative and M is polynomially large, so $\min_i y_i \ll -63$ and EI and PI are evaluated deep in their lower tails, where the argument says nothing. The common choice $f = \min_i y_i$ is therefore not covered. The incumbent $\min_i \mu(x_i)$, often used with noisy data, may be covered by the argument of Proposition 28, but we have not checked it.
- In the noise-free model, the observations are values of the posterior mean, and the gadget is positive at its own points. The next lemma and proposition show that $\min_i y_i$ is then about -1.

The positivity comes from evaluating the gadget at its own points. The second-order term Λ_E of Lemma 6 is at least 16 there, and the remainder is small because gadget points are close to each other.

Lemma 27 (The gadget at its own points)**.** *Assume $q \ge 1$, $\varepsilon_{\mathrm{m}} \le 0.01$ and $\Lambda \ge 2T_\star/\tau_0$. Then $\mu_\Lambda^*(p) \ge 7.3\,\gamma$ for every $p \in P$.*

Proof. We first show $\Lambda_E(p) \ge 16$ on P. At 0, $\pm e_i$ and $\pm\frac{1}{2}e_i$, at most one coordinate is nonzero, so $C(p) = 0$ and $\Lambda_E(p) = \beta_0 \ge 48$. At $p = ae_j + be_k$ with $\{j,k\} \in E$, the only edge with both coordinates nonzero is $\{j,k\}$, so $C(p) = ab$ and

$$\Lambda_E(p) \ge \beta_0 - 32 = 32\left(q - \frac{1}{2}\right) \ge 16.$$

Next, two gadget points each have at most two nonzero coordinates of size at most 1, so $\|p - p'\|^2 \le 8$. Lemma 6 and $\|\alpha^*\|_1 = 3/(4d^3\tau_0)$ then give, for $\tau \le \tau_0$ and $d \ge 5$,

$$|R_\tau(p)| \le \frac{8^3}{6}\|\alpha^*\|_1\tau^3 = \frac{64}{d^3}\cdot\frac{\tau^3}{\tau_0} \le 0.52\,\tau^2.$$

Hence $G_\tau(p) \ge 8\tau^2 - 0.52\tau^2 = 7.48\tau^2$ for $\tau \le \tau_0$, and $G_\tau(p) \ge -\|\alpha^*\|_1$ for $\tau > \tau_0$. Integrating and using Proposition 8 at $\ell^2 = \Lambda$,

$$\mu_\Lambda^*(p) \ge 7.48\,Q_\Lambda - E_\Lambda \ge (7.48 \cdot 0.99 - 0.01)\gamma \ge 7.3\gamma.$$ □

Proposition 28 (Data-determined incumbent, noise-free model)**.** *In the noise-free model, with $q \ge 1$, every gadget observation is positive, and in standardized units*

$$\min_i y_i \in [-1.04,\ -0.96].$$

In particular, expected improvement with incumbent $f = \min_i y_i$ is NP-hard.

Proof. Before rounding, a gadget observation is

$$(y_W)_p = \left[\Sigma_W(\theta^\circ)\bar{\alpha}\right]_p = M\mu^*_\Lambda(p) + J_{p,u_1}\beta_c,$$

because $\sigma_f^2 = 1$ and $\sigma_n = 0$ at the target. By Lemma 27, $M\mu^*_\Lambda(p) \geq 7.3\,M\gamma = 29.2$. By Lemma 12,

$$|J_{p,u_1}\beta_c| \leq \sqrt{n_1}\, g(4S/\Lambda) \leq n_{\max} g(4S/\Lambda) \leq 1.$$

Rounding moves the value by at most ρ, so every gadget observation is at least 28.
A calibration observation in block b of family f is $m_b + c_f$ plus its contrast terms. Here $m_b = \pm\xi_f$ with $\xi_f^2 \in [0.999, 1]$, $|c_f| \leq 0.004$, and the contrast terms are at most 0.02 in absolute value:

$$\zeta_h^2 \leq 1.01 m_1 h^2/\Lambda, \quad \zeta_a^2 \leq 1.01 m_1/\Lambda, \quad \zeta_{aa'} \leq 1.01\sqrt{m_2}/\Lambda,$$

as in Appendix C, with $\Lambda \geq 12800 m_1 + 10^4\sqrt{m_2}$. Since the signs of m_b alternate, some observation is at most $-0.9995 + 0.024 \leq -0.97$, and every observation is at least -1.024. Dividing by $c_s \in [0.99, 1.01]$ gives the stated range. The EI case of Corollary 26 covers every $f > -62.58$. □

Probability of improvement with $f = \min_i y_i$ is not covered in either model. In the noise-free model, μ interpolates the data, so $\min \mu \leq \min_i y_i$ in every instance, while the PI case needs $\min \mu > f$ in no instances.

Proposition 29 (Amplification)**.** *Let $c \geq 1$ be bounded by a polynomial in d. Replace M by cM, leave the calibration blocks unchanged, and use $\varepsilon_m = 0.01/c$ in Proposition 8 and $\varepsilon_w = 0.05$ in Lemma 20. Then the reduction stays polynomial, and in standardized units $\min \mu \leq -63c$ in yes instances and $\min \mu \geq -1.25$ in no instances. In particular, Corollary 26 holds with $\beta^{1/2} < (63c - 1.25)/1.03$, with $\sup h - \inf h < 63c - 1.25$ for the additive term and $\operatorname{osc}(c_0) < 63c - 1.25$ for the Gittins index, with $f > -63c + 0.42$ for EI, and with $f \in (-63c, -1.25)$ for PI.*

Proof. The hyperparameters are pinned by the calibration groups. The gadget enters the argument only through M, and we list each place.

- Proposition 8: $T_\star$ grows by the factor $c^{1/\omega}$, so Λ stays polynomial.
- Lemma 12: the condition becomes $B \geq (2n_0 + 2)^2 c^2 M^2 \|\alpha^*\|_1^2$.
- Lemma 20: Y and $A_{\max}$ grow by at most the factor c, and the conditions on η, ρ and ϵ_C shrink by at most c.
- Lemmas 18 and 47: Γ_0 contains $\frac{1}{2}\bar{\lambda}_W\|\bar{\alpha}\|^2$, which grows by c^2, so $B - 1 \geq 48\Gamma_1/\varepsilon_p^2$ grows by c^2.
- Condition (3) and the standardization condition

  $$B \geq 10^4 n_0 (n_0 + 2)^2 (1 + cM\|\alpha^*\|_1)^2$$

  hold with cM in place of M; S and B grow polynomially.
- Lemma 21(c): with $cM\gamma = 4c$ and $\varepsilon_m c = 0.01$, the yes bound becomes

  $$-0.98 \cdot 4c(24 - \varepsilon_m)/1.0404 + 0.051 \leq -90.3c,$$

  and the no bound stays

  $$-1.02(4c \cdot \varepsilon_m/0.9604 + 0.05) \geq -0.1.$$

- Lemma 21(b) and (d) do not involve M beyond (3), and the far-block coefficients of Appendix C are bounded in terms of Λ^2/m_2, not M.
- Lemma 31: the gadget part enters through $\eta\, cM$. The condition on η in Lemma 20 shrinks by the factor c, so (5) holds with cM in place of M, the weight errors are still at most $\varepsilon_{\mathrm{w}} = 0.05$, and the weight change is still at most $0.32\varepsilon_{\mathrm{w}} + 10^{-4}$. The near-block coefficients are bounded by $2.25 \cdot 4\Lambda^2/m_2 \le 2.25\, cM$, so the near-block change is still at most 0.0004. The family-sum part and the far blocks do not involve M beyond (3). The drift bound 0.065 is therefore unchanged.

The standardization ranges are unchanged, because the gadget share of the mean square is still at most 0.001. Dividing as in the proof of Theorem 3 gives the stated levels, and the thresholds follow from Lemma 23 as in Corollary 26. □

Theory-driven exploration weights such as those of Srinivas et al. (2010) grow with the dimension and the number of rounds. Proposition 29 covers any such weight bounded by a polynomial in d.

ε-optimality

Write $c_{\mathrm{s}} \in [0.99, 1.42]$ for the rational standardizing factor and $\theta^\circ_{\mathrm{s}}$ for θ° rescaled as in Lemma 4, so $\mu_{\theta^\circ_{\mathrm{s}}}(\cdot; y/c_{\mathrm{s}}) = \mu_{\theta^\circ}(\cdot; y)/c_{\mathrm{s}}$.

Lemma 30 (Evaluation)**.** *Given rational $x \in [-1,1]^D$, $\mu_{\theta^\circ_{\mathrm{s}}}(x)$ can be computed to within 0.05 in time polynomial in the instance size and the bit length of x.*

Proof. By Lemma 11,

$$\lambda_{\min}(\Sigma_{\theta^\circ}) \ge \lambda_* := \min\left\{\min_g \lambda_g(\theta^\circ), \frac{\lambda^-}{2}\right\},$$

where the second term bounds $\lambda_{\min}(\Sigma_W(\theta^\circ))$ as in the proof of Lemma 20. It follows that λ_* is 1/poly. By (A4), every entry of Σ_{θ° and $k_\Lambda(x)$ can be computed to any polynomial number of bits in polynomial time. If every entry of Σ_{θ° is perturbed by at most δ_E with $n\delta_E \le \lambda_*/2$, the solution of $\Sigma_{\theta^\circ}\alpha = y$ moves by at most $2n\delta_E\|y\|/\lambda_*^2$, and since $\|k(x)\| \le \sqrt{n}$ the value $\sigma_f^2 k(x)^\top \Sigma^{-1} y$ moves by at most a polynomial multiple of δ_E. Choosing $\delta_E = 2^{-p}$ for a suitable polynomial p makes the error at most 0.04. Solving the rational system exactly by Gaussian elimination takes polynomial time, and dividing by $c_{\mathrm{s}} \ge 0.99$ keeps the error below 0.05. □

Lemma 31 (Stability across the window)**.** *For every $\theta \in \mathcal{W}_\eta$ and every $x \in [-1,1]^D$, $|\mu_\theta(x) - \mu_{\theta^\circ}(x)| \le 0.065$ in unstandardized units, and hence $|\mu_{\hat\theta}(x) - \mu_{\theta^\circ_{\mathrm{s}}}(x)| \le 0.066$ in standardized units for every joint MLE $\hat\theta$.*

Proof. Decompose $\mu_\theta = \mu_{\mathrm{gad}} + \mu_u + \mu_{\mathrm{grp}}$ as in Section 7, and likewise at $\theta^\circ \in \mathcal{W}_\eta$. Bounding each part separately at both endpoints, as Lemma 21 does, would charge the near-block between term of Appendix C, which is about 1, twice. Instead we bound the change of each part. Two facts are used throughout. First, by the choice of η in Lemma 20,

$$\eta M\|\alpha^*\|_1 \le \eta M\sqrt{n_0}\|\alpha^*\|_2 \le \frac{0.05\,\lambda^-}{5n_0} \le 0.001, \qquad \text{and } \|\alpha^*\|_1 \ge 4\beta_0 \ge 64, \tag{5}$$

so $\eta M \le 1.6 \cdot 10^{-5}$. Second, $|k_\ell - k_\Lambda| \le 0.38\eta$ pointwise by Lemma 7(e).

Gadget part. Write $\alpha_g = (\alpha_W)_{\text{gad}}$ at θ and α_g° at θ°. The first block equation of Lemma 20 has a right side that does not depend on θ, so

$$A_\theta(\alpha_g - \alpha_g^\circ) = (A_{\theta^\circ} - A_\theta)\alpha_g^\circ + J_{\theta^\circ}\alpha_u^\circ - J_\theta\alpha_u.$$

By Lemma 20, $\|\alpha_g^\circ\|_2 \le M\|\alpha^*\|_2 + 0.05$ and $\|\alpha_u\|, \|\alpha_u^\circ\| \le |\beta_c| + 0.5 \le 1.5$. Lemma 19 then gives

$$\|\alpha_g - \alpha_g^\circ\|_1 \le \frac{\sqrt{n_0}}{\lambda^-}\Big(1.6 n_0 \eta\,(M\|\alpha^*\|_2 + 0.05) + 3\epsilon_C\Big) \le 0.32\,\varepsilon_{\text{w}} + 10^{-4} \le 0.0161,$$

using the conditions on η and ϵ_C in Lemma 20 (the last two contributions carry an extra factor at most $1/(M\|\alpha^*\|_2)$). With $p' = (p, -1)$,

$$\mu_{\text{gad},\theta} - \mu_{\text{gad},\theta^\circ} = \sigma_f^2 \sum_p (\alpha_g - \alpha_g^\circ)_p k_\ell(x, p') + (\sigma_f^2 - 1)\sum_p (\alpha_g^\circ)_p k_\ell(x, p') + \sum_p (\alpha_g^\circ)_p (k_\ell - k_\Lambda)(x, p'),$$

and $\|\alpha_g^\circ\|_1 \le M\|\alpha^*\|_1 + 0.05$, so by (5) the change is at most $1.02 \cdot 0.0161 + 1.38\,(0.001 + 0.05\eta) \le 0.018$.

Family-sum part. By the proof of Lemma 21(b), $|\mu_u| \le 6.12 \cdot 0.003 \le 0.0184$ at each endpoint, so it changes by at most 0.037.

Group part: far blocks. Fix x and its near block as in Appendix C. The far blocks contribute at most 0.003 at each endpoint, so their change is at most 0.006.

Group part: near block. The near block has at most four distinct points x_j, each replicated r times. Its contribution is $\sum_j A_j(\theta)\, k_\ell(x, x_j)$, where

$$A_j(\theta) = \sum_g (y_g)_{x_j}\, \varrho_g(\theta), \qquad \varrho_g(\theta) := \frac{r\sigma_f^2}{\lambda_g(\theta)} = \frac{1}{\kappa_g(\ell) + \sigma_n^2/(r\sigma_f^2)},$$

the sum running over the at most four groups that involve the near block. We claim:

(i) $\sum_g |(y_g)_{x_j}|\, \varrho_g(\theta^\circ) \le 2.25M$ for each j;

(ii) $|\varrho_g(\theta)/\varrho_g(\theta^\circ) - 1| \le 2.2\eta$ for each such group.

Given these,

$$\begin{aligned}\left|\sum_j A_j(\theta) k_\ell(x, x_j) - \sum_j A_j(\theta^\circ) k_\Lambda(x, x_j)\right| &\le \sum_j \big(|A_j(\theta) - A_j(\theta^\circ)| + |A_j(\theta^\circ)|\, 0.38\eta\big) \\ &\le 4 \cdot 2.25M\,(2.2 + 0.38)\,\eta \\ &\le 0.0004\end{aligned}$$

by (5).

For (i), in the noise-free model ($r = 1$) this is the far-coefficient bound $9\Lambda^2/m_2 = 2.25M$ of Appendix C. In the noisy model, the within term is at most $\zeta_h \ell^2/(m_1 h^2) \le 2.02\sqrt{\Lambda/m_1}$, by the bounds on $\lambda_{P_h\text{-within}}$ and ζ_h in Appendix C, and the between term is at most $1.0001/1.99 \le 0.51$. Their sum is at most $3\sqrt{\Lambda/m_1}$. Using $m_2 \ge m_1^2$ (Jensen) and $\Lambda \ge 16 m_2/m_1$,

$$\frac{M}{\sqrt{\Lambda/m_1}} = \frac{4\Lambda^{3/2}\sqrt{m_1}}{m_2} \ge 4\Big(\frac{16 m_2}{m_1}\Big)^{3/2}\frac{\sqrt{m_1}}{m_2} = \frac{256\sqrt{m_2}}{m_1} \ge 256,$$

so $3\sqrt{\Lambda/m_1} \le M/85 \le 2.25M$.

For (ii), define

$$c = \Lambda/\ell^2 \in [(1+\eta)^{-1}, (1-\eta)^{-1}].$$

Since F is concave and increasing with $F(0) = 0$, $F(cv)/F(v)$ lies between 1 and c, so every within and pattern-a eigenvalue $\kappa_g = F(\cdot)$ changes by a relative amount at most 1.03η. For pattern aa', $\kappa_g = \int (1 - e^{-tv})^2 \, \mathrm{d}\pi$, and $1 - e^{-cu}$ lies between $1 - e^{-u}$ and $c(1 - e^{-u})$, so the relative change is at most $c^2 - 1 \le 2.1\eta$. The between eigenvalues are at least 1.99 and are combinations of at most seven kernel values with coefficients $\pm 1, \pm 2, \pm 4$ summing in absolute value to at most 8, so their relative change is at most $8 \cdot 0.38\eta/1.99 \le 1.6\eta$. Finally, $\sigma_n^2/(r\sigma_f^2)$ changes by a relative amount at most $(1+\eta)/(1-\eta) - 1 \le 2.05\eta$, and it is 0 in the noise-free model. A sum of two positive terms changes by at most the larger relative change, so $1/\varrho_g$ changes by a relative amount at most 2.1η, and ϱ_g by at most $2.1\eta/(1 - 2.1\eta) \le 2.2\eta$.

Total. Adding the four parts,

$$|\mu_\theta(x) - \mu_{\theta^\circ}(x)| \le 0.018 + 0.037 + 0.006 + 0.0004 \le 0.065.$$

For the standardized statement, every joint MLE lies in $\mathcal{W}_\eta$ (Proposition 14), and by Lemma 4 the standardized posterior means are the unstandardized ones divided by $c_\mathrm{s} \ge 0.99$, and $0.065/0.99 \le 0.066$. □

The drift is governed by the parts that are bounded at each endpoint rather than differenced: the family-sum part and the far blocks. Everything that depends on θ through the pinned hyperparameters moves by $O(\eta M)$, which the choice of η makes negligible.

Corollary 32 (ε-optimality)**.** *Unless* P = NP*, no polynomial-time algorithm, given an instance as in Theorem 3, does any of the following.*

(a) *Outputs a number within ε of $\min \mu_{\hat\theta}$, for any $\varepsilon < 30$.*

(b) *With θ_s° supplied, outputs a rational $\hat x$ with $\mu_{\theta_\mathrm{s}^\circ}(\hat x) \le \min \mu_{\theta_\mathrm{s}^\circ} + \varepsilon$, for any $\varepsilon < 61.6$.*

(c) *With fitted hyperparameters, outputs a rational $\hat x$ with $\mu_{\hat\theta}(\hat x) \le \min \mu_{\hat\theta} + \varepsilon$, for any $\varepsilon < 61.5$, in both models.*

(d) *For A as in Lemma 23 with gap $\Delta_a = a_+ - a_-$, outputs a value within ε of $\min A$ for any $\varepsilon < \Delta_a/2$; or, with θ_s° supplied, a point $\hat x$ with $A(\hat x) \le \min A + \varepsilon$ for any $\varepsilon < \inf_{s \in [0, \bar s]} a(-1.35, s) - a_-$.*

Proof. In each part such an algorithm would decide Max-Cut. The bounds $\min \mu \le -63$ (yes) and $\min \mu \ge -1.25$ (no) hold for every $\theta \in \mathcal{W}_\eta$, not only at $\hat\theta$, because the proof of Theorem 3 fixes an arbitrary θ in the window; in particular they hold at θ_s°.

(a) If $\hat v$ is the output, then $\hat v \le -63 + \varepsilon < -32.1$ in yes instances and $\hat v \ge -1.25 - \varepsilon > -32.1$ in no instances.

(b) The output $\hat x$ has polynomial bit length. Compute $\tilde v$, an approximation of $\mu_{\theta_\mathrm{s}^\circ}(\hat x)$ to within 0.05 (Lemma 30). In yes instances $\tilde v \le -63 + \varepsilon + 0.05 < -1.3$ when $\varepsilon < 61.6$; in no instances $\tilde v \ge -1.3$.

(c) As in (b), but the output is near-optimal for $\mu_{\hat\theta}$, and the algorithm sees only the data. The reduction, not the algorithm, knows θ_s°, and uses it only to check the output. By Lemma 31, divided by $c_\mathrm{s} \ge 0.99$, $\mu_{\theta_\mathrm{s}^\circ}(\hat x) \le -63 + \varepsilon + 0.066$ in yes instances, so $\tilde v \le -62.88 + \varepsilon < -1.3$ when $\varepsilon < 61.5$. In no instances $\tilde v \ge -1.3$.

(d) By Lemma 23, $\min A \le a_-$ in yes instances and $\min A \ge a_+$ in no instances, at every $\theta \in \mathcal{W}_\eta$. For values, compare the output with $(a_- + a_+)/2$. For points, compute $\tilde{v}$, an approximation of $\mu_{\theta_s^\circ}(\hat{x})$ to within 0.05, and answer "yes" if and only if $\tilde{v} < -1.3$. In no instances $\tilde{v} \ge -1.3$. In yes instances $A(\hat{x}) \le a_- + \varepsilon < \inf_s a(-1.35, s)$, so monotonicity in the first argument forces $\mu_{\theta_s^\circ}(\hat{x}) < -1.35$, and $\tilde{v} < -1.3$. This needs only Lemma 30, not an evaluation of a. □

The reduction in Corollary 32(c) never computes $\hat{\theta}$; it only checks the algorithm's output against $\mu_{\theta_s^\circ}$. The same check works for any acquisition function covered by Lemma 23. Monotonicity in μ converts near-optimality of $A_{\hat{\theta}}$ at $\hat{x}$ into an upper bound on $\mu_{\hat{\theta}}(\hat{x})$, and Lemma 31 carries that bound to θ_s°. Write

$$c_{\text{fit}} := 1.42 \;\ge\; c_{\text{no}} + 0.066 + 0.1,$$

where 0.066 is the drift of Lemma 31 in standardized units and 0.1 covers the evaluation error of Lemma 30 on both sides of the threshold.

Corollary 33 (Acquisition functions with fitted hyperparameters: points)**.** *Let $a\colon \mathbb{R} \times [0, \bar{s}] \to \mathbb{R}$ be nondecreasing in its first argument, let $A_\theta(x) = a(\mu_\theta(x), \sigma_\theta(x))$, and let*

$$a_- = \sup_{s \in [0,\bar{s}]} a(-c_{\text{yes}}, s), \qquad a'_+ = \inf_{s \in [0,\bar{s}]} a(-c_{\text{fit}}, s), \qquad \Delta'_a = a'_+ - a_-.$$

Unless P = NP*, no polynomial-time algorithm, given an instance as in Theorem 3, outputs a rational $\hat{x} \in [-1, 1]^D$ with*

$$A_{\hat{\theta}}(\hat{x}) \le \min_{[-1,1]^D} A_{\hat{\theta}} + \varepsilon$$

for some joint MLE $\hat{\theta}$, for any $\varepsilon < \Delta'_a$. This holds in both models. The algorithm sees only the data; neither the reduction nor the check needs $\hat{\theta}$, σ, or an evaluation of a.

Proof. Since $c_{\text{fit}} > c_{\text{no}}$ and a is nondecreasing, $a'_+ \le a_+$, so $\Delta'_a \le \Delta_a$. The output $\hat{x}$ has polynomial bit length. Compute $\tilde{v}$, an approximation of $\mu_{\theta_s^\circ}(\hat{x})$ to within 0.05 (Lemma 30), and answer "yes" if and only if $\tilde{v} < -1.3$.

No instances. θ_s° lies in the window, so $\mu_{\theta_s^\circ} \ge -c_{\text{no}} = -1.25$ on $[-1, 1]^D$, and $\tilde{v} \ge -1.3$.

Yes instances. Every joint MLE lies in the window, so Lemma 23 applies at $\hat{\theta}$ and gives $\min A_{\hat{\theta}} \le a_-$. Hence

$$A_{\hat{\theta}}(\hat{x}) \le a_- + \varepsilon < a'_+.$$

Suppose $\mu_{\hat{\theta}}(\hat{x}) \ge -c_{\text{fit}}$. Since $\sigma_{\hat{\theta}}(\hat{x}) \in [0, \bar{s}]$ by Lemma 22 and a is nondecreasing in its first argument,

$$A_{\hat{\theta}}(\hat{x}) = a\big(\mu_{\hat{\theta}}(\hat{x}), \sigma_{\hat{\theta}}(\hat{x})\big) \ge a\big(-c_{\text{fit}}, \sigma_{\hat{\theta}}(\hat{x})\big) \ge a'_+,$$

a contradiction. Therefore $\mu_{\hat{\theta}}(\hat{x}) < -c_{\text{fit}}$. By Lemma 31, divided by $c_s \ge 0.99$,

$$\mu_{\theta_s^\circ}(\hat{x}) < -c_{\text{fit}} + 0.066 < -1.35,$$

so $\tilde{v} < -1.3$. The test therefore decides Max-Cut. □

The constants enter only through c_{fit}. Compared with Corollary 32(d), the price of not knowing $\hat{\theta}$ is that the level -1.35 there is replaced by $-c_{\text{fit}} = -1.42$, a loss of only 0.07. Corollary 32(c) is the case $a(m, s) = m$. For the acquisition functions of Corollary 26, the gap is bounded as follows. For expected improvement and probability of improvement, which are maximized and nonincreasing in μ, apply the corollary to $a = -\text{EI}$ and $a = -\text{PI}$; an ε-optimal point for $\max \text{EI}$ is an ε-optimal point for $\min(-\text{EI})$.

Acquisition	Covered when	ε-optimal points hard for $\varepsilon <$
LCB, $\mu - \beta^{1/2}\sigma$	$\beta^{1/2} < 59.7$	$61.58 - 1.03\,\beta^{1/2}$
Additive σ-term	$\sup h - \inf h < 61.58$	$61.58 - (\sup h - \inf h)$
Gittins index	$\operatorname{osc}(c_0) < 61.58$; e.g., $c_0 \geq 10^{-760}$	$61.58 - \operatorname{osc}(c_0)$
EI, incumbent f	$f > -62.58$	$f + 62.58 - \max\{f + 1.42, 0\}$
EI, incumbent $\min_i y_i$ (noise-free)	always	61.16
PI, incumbent f	$f \in (-63, -1.42)$	$\Phi\big(\frac{f+63}{\bar{s}}\big) - \Phi\big(\frac{f+1.42}{\bar{s}}\big)$

Derivations.

- LCB: $a_- = -63$ (at $s = 0$) and $a'_+ = -1.42 - 1.03\,\beta^{1/2}$.
- Additive term: $a_- = -63 - \inf h$ and $a'_+ = -1.42 - \sup h$.
- Gittins: $a_- = -63 + c_0$ and $a'_+ = -1.42 + h(\bar{s})$, so the gap is $61.58 - \operatorname{osc}(c_0)$. By Lemma 25, the gap exceeds 61.4 for every $c_0 \geq 1$, is about 32.2 at $c_0 = 10^{-180}$, and stays positive at least down to $c_0 = 10^{-760}$. Remark 34 covers smaller costs.
- EI: by (4), $-\mathrm{EI}(-63, s) \leq -(f + 63)$ for every s, and

$$-\mathrm{EI}(-1.42, s) \geq -\max\{f + 1.42, 0\} - 1.03\,\phi(0) \geq -\max\{f + 1.42, 0\} - 0.42.$$

  For $f \geq -1.42$ the gap is 61.16. The incumbent $\min_i y_i$ is a function of the data alone, so in the noise-free model it lies in $[-1.04, -0.96]$ (Proposition 28) whatever $\hat{\theta}$ is.
- PI: as in Corollary 26 with -1.25 replaced by -1.42; at $f = -32$ the gap is $1 - \epsilon$ with $\epsilon < 10^{-100}$.

Remark 34 (Amplification)**.** Under Proposition 29, the yes level becomes $-63c$ while c_{no}, the drift 0.066 of Lemma 31 (see the additional item in the proof of Proposition 29) and hence c_{fit} are unchanged. Corollary 33 then holds with $a_- = \sup_s a(-63c, s)$. For LCB this gives hardness for $\varepsilon < 63c - 1.42 - 1.03\,\beta^{1/2}$, and for the Gittins index for $\varepsilon < 63c - 1.42 - \operatorname{osc}(c_0)$. So every polynomially bounded exploration weight is covered with a gap that grows with c, and so is every Gittins cost c_0 of polynomial bit length, since $\operatorname{osc}(c_0) \leq \sqrt{\mathrm{poly}}$ by Lemma 25. (A bound in terms of $\sup|h|$ would not suffice here: for $c_0 = 2^{\mathrm{poly}}$ it equals c_0, while $\operatorname{osc}(c_0) \to 0$.)

Remark 35 (Values)**.** For values, Corollary 32(d) already applies with fitted hyperparameters, because Lemma 23 holds at every $\theta \in \mathcal{W}_\eta$. The tolerance must be strictly less than $\Delta_a/2$: with equality, an output equal to $(a_- + a_+)/2$ is consistent with both cases.

Hardness at one fixed ε implies hardness at every smaller ε.

9 Fixed dimension is easy with supplied hyperparameters

The instances of Theorems 2 and 3 have dimension growing polynomially with the size of the graph. This section shows that some growth is necessary: in fixed dimension, a grid decides the constant-gap problem in polynomial time. Fix any θ, and write the posterior mean as

$$\mu_\theta = \sum_i (\alpha_\theta)_i\, k_\ell(\cdot, x_i), \qquad \alpha_\theta = \sigma_f^2 \Sigma_\theta^{-1} y.$$

Let $\hat{H}$ be its norm in the reproducing-kernel Hilbert space of the unit-variance kernel k_ℓ:

$$\hat{H}^2 := \alpha_\theta^\top K_\ell \, \alpha_\theta.$$

It can be computed in $O(n^2)$ operations once α_θ is known.

Lemma 36 (Values and slope)**.** *Assume (A1). For every θ and all $x, x' \in \mathbb{R}^D$,*

$$|\mu_\theta(x)| \le \hat{H}, \qquad |\mu_\theta(x) - \mu_\theta(x')| \le \hat{H}\sqrt{2m_1}\,\frac{\|x - x'\|}{\ell}.$$

Proof. By the reproducing property and Cauchy–Schwarz,

$$|\mu_\theta(x) - \mu_\theta(x')| \le \hat{H}\,\|k_\ell(\cdot, x) - k_\ell(\cdot, x')\| = \hat{H}\sqrt{2F(\|x - x'\|^2/\ell^2)},$$

and $F(v) \le m_1 v$ by Lemma 7(a). The first bound is the same argument with $\|k_\ell(\cdot, x)\| = 1$. □

Lemma 37 ($\hat{H}$ at a likelihood maximum)**.** *Suppose $\sigma_f > 0$ and θ is stationary for L along the ray*

$$c \mapsto \theta_c := (\ell^2, c\,\sigma_f^2, c\,\sigma_n^2),$$

as at any joint MLE and at any local maximizer. Then $y^\top \Sigma_\theta^{-1} y = n$, and

$$\hat{H}^2 = \sigma_f^2\big(n - \sigma_n^2 \|\Sigma_\theta^{-1} y\|^2\big) \le n\sigma_f^2.$$

If instead σ_n^2 is held at a lower bound and θ maximizes L subject to it, then $y^\top \Sigma_\theta^{-1} y \le n$ and still $\hat{H}^2 \le n\sigma_f^2$.

Proof. Since $\Sigma_{\theta_c} = c\,\Sigma_\theta$,

$$L(\theta_c) = -\frac{1}{2}\Big[n \log c + \log\det \Sigma_\theta + \frac{y^\top \Sigma_\theta^{-1} y}{c}\Big],$$

so the derivative at $c = 1$ is $-\frac{1}{2}(n - y^\top \Sigma_\theta^{-1} y)$. It vanishes at a stationary point. At a lower bound on σ_n^2 only $c \ge 1$ is feasible, so the derivative is at most 0, which gives $y^\top \Sigma_\theta^{-1} y \le n$. Finally, $K_\ell = (\Sigma_\theta - \sigma_n^2 I)/\sigma_f^2$ gives

$$\alpha_\theta^\top K_\ell \, \alpha_\theta = \sigma_f^2\, y^\top \Sigma_\theta^{-1}(\Sigma_\theta - \sigma_n^2 I)\Sigma_\theta^{-1} y = \sigma_f^2\big(y^\top \Sigma_\theta^{-1} y - \sigma_n^2 \|\Sigma_\theta^{-1} y\|^2\big).$$ □

In the noise-free model, $\hat{\sigma}_f^2 = y^\top K_\ell^{-1} y/n$ is the classical profile estimate of kriging (Sacks et al., 1989; Jones et al., 1998), and $\hat{H}^2 = y^\top K_\ell^{-1} y$ is the squared native-space norm of the interpolant (Wendland, 2004).

For an integer $N \ge 1$, let $G_N = \{-1, -1 + \Delta, \dots, 1\}^D$ with $\Delta = 2/N$. It has $(N+1)^D$ points.

Theorem 38 (Grid)**.** *Assume (A1), and let $\varepsilon > 0$. If*

$$N \ge \frac{\sqrt{2m_1 D}\,\hat{H}}{\ell\,\varepsilon},$$

then $\min_{G_N} \mu_\theta \le \min_{[-1,1]^D} \mu_\theta + \varepsilon$. Computing α_θ and evaluating μ_θ on G_N takes

$$O\big(n^3 + nD(N+1)^D\big)$$

arithmetic operations and kernel evaluations.

Proof. Let x^* be a minimizer. Some grid point z satisfies $\|z-x^*\| \le \frac{\Delta}{2}\sqrt{D} = \sqrt{D}/N$, and Lemma 36 gives $\mu_\theta(z) - \mu_\theta(x^*) \le \varepsilon$. Solving for α_θ takes $O(n^3)$ operations, and each evaluation of μ_θ takes $O(nD)$. □

Proposition 39 (Fixed dimension is easy)**.** *Consider the noisy model with θ supplied, and constants $c_a > c_b$. Deciding whether $\min \mu_\theta \le -c_a$ or $\min \mu_\theta \ge -c_b$ takes time polynomial in the input size and in*

$$\left(2 + \frac{\sqrt{D}\,\hat{H}}{\ell}\right)^{O(D)},$$

with a constant depending on $c_a - c_b$ and g. Hence the problem is in P when D is fixed and $\hat{H}/\ell$ is polynomially bounded. It is in quasi-polynomial time when $D \le (\log n)^k$ and $\hat{H}/\ell \le 2^{(\log n)^k}$ for a constant k, so it is not NP-hard on such instances unless NP has quasi-polynomial-time algorithms.

Proof. Run the grid of Theorem 38 with $\varepsilon = (c_a - c_b)/3$, evaluating μ_θ to within $\varepsilon/3$, and compare the grid minimum with $-(c_a+c_b)/2$. Since $\lambda_{\min}(\Sigma_\theta) \ge \sigma_n^2$, the precision needed for α_θ is polynomial in the bit length of θ and y, as in Lemma 30. □

On the instances of Theorem 2, $\hat{H} \le \sqrt{n_0}\,(M\|\alpha^*\|_1 + 1)$ and $\ell^2 = \Lambda \ge 1$, so $\hat{H}/\ell$ is polynomial. In the noisy model, their hardness therefore requires D to grow. At a joint MLE, Lemma 37 gives $\hat{H} \le \sqrt{n}\,\hat{\sigma}_f$, and on the instances of Theorem 3, $\hat{\sigma}_f^2 \le 1.02$ and $\hat{\ell}^2 \ge 0.98\Lambda \ge 1$ before standardization. So the posterior means at a joint MLE also have polynomially bounded $\hat{H}/\ell$. This alone does not make fixed dimension easy with fitted hyperparameters, since an algorithm would also have to compute a joint MLE; Proposition 39 concerns supplied hyperparameters only. The noise-free model needs one more assumption, $\lambda_{\min}(K_\ell) \ge 2^{-\mathrm{poly}}$, because for the squared-exponential kernel $\log(1/\lambda_{\min}(K_\ell))$ can grow like ℓ^2/s^2, where s is the smallest distance between two inputs.

Remark 40 (Why the calibration uses many coordinates)**.** Suppose a design contains B congruent blocks with distinct centroids, and every pair of blocks has the same multiset of cross-distances, as in Section 4. Then the centroids are pairwise equidistant. Pairwise equidistant points form a regular simplex, so $D \ge B - 1$. Exact invariant subspaces of this kind therefore cost at least one dimension per block. Our design uses $S+4$ coordinates per block, because the hubs must also make far-away blocks negligible. A reduction in lower dimension would need approximate symmetry, and the pinning argument would need a perturbation version.

10 Concluding remarks

Hardness follows from pairing an isotropic kernel that is a scale mixture of Gaussians, with slightly more than two finite moments, with an acquisition rule that is monotone in μ and moves by less than the gap as σ ranges over $[0, \bar{s}]$. It survives standardization and maximum-likelihood fitting of every hyperparameter, with or without noise. This is the Gaussian-process analogue of the NP-hardness of indefinite quadratic minimization over a box (Sahni, 1974; Pardalos and Vavasis, 1991); here the extreme-point step is the multilinearity of the cut polynomial in Lemma 6.

Limitations. Our results assume an isotropic kernel with zero prior mean. Theorem 3 assumes a global maximizer of the likelihood, but by the remark after Proposition 14 it also covers any local maximizer reached by ascent from θ_s°, the target rescaled to the standardized data. By Proposition 15, it also covers MAP estimates under any bounded continuous hyperprior, at the cost

of a polylogarithmic increase in B for standard priors. The incumbent in EI and PI is supplied, except for EI in the noise-free model (Proposition 28), and it must lie in the ranges of Corollary 26; for PI, between -63 and -1.25 in standardized units. The constructed instances have dimension polynomial in n. Section 9 shows that some growth is necessary when the hyperparameters are supplied: in the noisy model, with $\hat{H}/\ell$ polynomially bounded, dimension $D \le (\log n)^k$ gives a quasi-polynomial-time algorithm, but we do not know whether dimension polynomial in n is needed. The posterior mean of these instances also dips far below the range of the fitted prior: the yes-instance minimum lies at least $c_{\text{yes}}/\bar{s} \ge 61$ fitted prior standard deviations below the prior mean, at a point where $\sigma \le \bar{s}$. The likelihood tolerates this because the calibration data, almost all of the n observations, set $\hat{\sigma}_f$, while the gadget is a small cluster of large norm whose cost, $\Gamma_1 \ni \frac{1}{2}\bar{\lambda}_W \|\bar{\alpha}\|^2 \propto M^2$, is paid for by the size of B. The supplied incumbents covered by Corollary 26 are much less extreme: for objectives with sharp optima, standardized running minima below -1.25 are common at moderate budgets.

Open problems.

- Hardness for kernels whose mixing measure has an infinite second moment, such as the Matérn kernels with $\nu \le 2$, including the widely used $\nu = \frac{3}{2}$. The moment-matched gadget needs $m_2 < \infty$, since its margin is the second-order term, and our proof needs slightly more, $m_{2+\omega} < \infty$.
- Hardness with automatic relevance determination, that is, one length scale per coordinate. Each length scale would need its own pinning family, so the dimension would grow with the number of distinct length scales.
- Hardness with a fitted constant prior mean.
- Hardness with data-determined incumbents in the noisy model, and probability of improvement with $f = \min_i y_i$ in either model.
- Hardness on self-consistent instances, for example with $\min \mu \ge -C\hat{\sigma}_f$ for a small constant C. This needs a calibration design without the floor of about -1 that comes from the block means.
- Intermediate dimension, $(\log n)^{\omega(1)} \le D \le n^{o(1)}$, and the complexity of computing a joint MLE in fixed dimension.

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A Verifying the identifiability assumption (A3)

For a positive C^1 function φ on $(0,\infty)$, its elasticity is

$$\epsilon_\varphi(u) = \frac{u\varphi'(u)}{\varphi(u)} = \frac{\mathrm{d}\log\varphi}{\mathrm{d}\log u}.$$

Reparametrizing by $u' = cu^p$ with $c, p > 0$ multiplies the elasticity by p, so whether ϵ_φ is strictly decreasing does not depend on which power of the distance is used as the variable.

Lemma 41 (Elasticity criterion)**.** *If ϵ_F is strictly decreasing on $(0,\infty)$, then (A3) holds.*

Proof. $\frac{\mathrm{d}}{\mathrm{d}v}\log\mathcal{R}(v) = \frac{4F'(4v)}{F(4v)} - \frac{F'(v)}{F(v)} = \frac{\epsilon_F(4v)-\epsilon_F(v)}{v} < 0.$ □

Lemma 42 (Inheritance)**.** *Let $f > 0$ be C^1 on $(0,\infty)$ and integrable at 0, and $F(u) = \int_0^u f$. If ϵ_f is strictly decreasing, so is ϵ_F.*

Proof. Substituting $y = tu$,

$$\epsilon_F(u) = \frac{uf(u)}{\int_0^u f(y)\,\mathrm{d}y} = \left(\int_0^1 \frac{f(tu)}{f(u)}\,\mathrm{d}t\right)^{-1}.$$

For fixed $t \in (0,1)$, $\frac{\mathrm{d}}{\mathrm{d}u}\log\frac{f(tu)}{f(u)} = \frac{\epsilon_f(tu)-\epsilon_f(u)}{u} > 0$ because $tu < u$, so every integrand is strictly increasing in u. The integral equals $F(u)/(uf(u))$, which is finite, so it is strictly increasing and ϵ_F is strictly decreasing. □

The above implies that (A3) holds whenever $-g'$ has strictly decreasing elasticity; that is, whenever $s \mapsto \log(-g'(e^s))$ is strictly concave. This is immediate for the squared exponential kernel with

$$g = e^{-u/2}, \quad \epsilon_{-g'}(u) = -u/2;$$

the rational quadratic kernels (including the generalized Cauchy kernel with $\varsigma = 2$) with

$$g = (1+u/c)^{-\alpha}, \quad \epsilon_{-g'}(u) = -(\alpha+1)\frac{u/c}{1+u/c};$$

and the power-exponential kernels with

$$g = \exp(-u^{p/2}), \quad 0 < p \le 2, \quad \epsilon_{-g'}(u) = \frac{p}{2} - 1 - \frac{p}{2}u^{p/2}.$$

For $p < 2$, however, $m_1 = \infty$, so (A1) fails and these kernels are not covered by Theorems 2 and 3. For the Matérn family, $-g'$ is a power times a Bessel function $K_{\nu-1}$, and its elasticity is governed by the ratio $K_\nu/K_{\nu-1}$.

Lemma 43 (Bessel ratio)**.** *For real υ and $x > 0$, let $\varpi(x) = K_{\upsilon+1}(x)/K_\upsilon(x)$. Then*

$$\varpi(x) > \underline{\varpi}(x) := \frac{\upsilon + \sqrt{\upsilon^2 + x^2}}{x},$$

and $x \mapsto x\varpi(x)$ is strictly increasing on $\mathbb{R}_+$.

Bounds of this type appear in Segura (2022). We include a proof of both claims for every real υ.

Proof. Because $K_\upsilon' = -K_{\upsilon+1} + \frac{\upsilon}{x}K_\upsilon$ and $K_{\upsilon+1}' = -K_\upsilon - \frac{\upsilon+1}{x}K_{\upsilon+1}$, the quotient rule gives

$$\varpi' = \varpi^2 - \frac{2\upsilon+1}{x}\varpi - 1.$$

$\underline{\varpi}$ is the positive root of $z^2 - \frac{2\upsilon}{x}z - 1 = 0$, so at any point where $\varpi = \underline{\varpi}$,

$$\varpi' = \underline{\varpi}^2 - \tfrac{2\upsilon+1}{x}\underline{\varpi} - 1 = -\underline{\varpi}/x.$$

Since $\underline{\varpi}' = -\upsilon\underline{\varpi}/(x\sqrt{\upsilon^2+x^2})$, the difference $\mathcal{D} := \varpi - \underline{\varpi}$ satisfies

$$\mathcal{D}' = -\frac{\varpi}{x}\Big(1 - \frac{\upsilon}{\sqrt{\upsilon^2+x^2}}\Big) < 0$$

at every zero, so $\mathcal{D}$ can only cross zero downward. From

$$K_\upsilon(x) = \sqrt{\pi/(2x)}\,e^{-x}\left(1 + \frac{4\upsilon^2-1}{8x} + O(x^{-2})\right)$$

we get $\varpi = 1 + \frac{2\upsilon+1}{2x} + O(x^{-2})$ and $\underline{\varpi} = 1 + \frac{\upsilon}{x} + O(x^{-2})$, so $\mathcal{D} = \frac{1}{2x} + O(x^{-2}) > 0$ for large x. A function that is positive for large x and can only cross zero downward is positive everywhere, so $\varpi > \underline{\varpi}$. Finally

$$(x\varpi)' = \varpi + x\varpi' = x\Big(\varpi^2 - \tfrac{2\upsilon}{x}\varpi - 1\Big) = x(\varpi - \underline{\varpi})(\varpi - \varpi_-), \qquad \varpi_- = \frac{\upsilon - \sqrt{\upsilon^2+x^2}}{x} < 0 < \varpi,$$

and both factors are positive. □

Proposition 44 (Matérn)**.** *For every $\nu > 0$, the Matérn profile $g_\nu(z) = \frac{2^{1-\nu}}{\Gamma(\nu)}z^\nu K_\nu(z)$, with $z = \sqrt{2\nu u}$, satisfies (A3).*

Proof. The identity $(z^\nu K_\nu)' = -z^\nu K_{\nu-1}$ gives

$$f(z) := -g_\nu'(z) = \frac{2^{1-\nu}}{\Gamma(\nu)}z^\nu K_{\nu-1}(z) > 0.$$

It is integrable at 0, and $F(z) = \int_0^z f = 1 - g_\nu(z)$ because $g_\nu(0^+) = 1$. Using $K_{\nu-1}' = -K_\nu + \frac{\nu-1}{z}K_{\nu-1}$,

$$\epsilon_f(z) = \nu + \frac{zK_{\nu-1}'(z)}{K_{\nu-1}(z)} = 2\nu - 1 - z\,\varpi_{\nu-1}(z),$$

where $\varpi_{\nu-1}$ is the ratio of Lemma 43 with $\upsilon = \nu - 1$. This is strictly decreasing by Lemma 43. Lemmas 42 and 41 finish the proof; by the reparametrization remark, using z rather than u as the variable is harmless. □

This covers every Matérn smoothness, including $\nu \le 2$, which is excluded from Theorem 3 only because (A1) fails there.

B The noise-free model

Throughout this appendix, the model is noise-free, $r = 1$, $\mathcal{F} = \{P_1, P_2, \mathrm{Sq}\}$ and $n_{\max} = 4B$. The first part checks (H1) and (H2); the rest proves the spectral floor.

Lemma 45 (Noise-free spectral floor)**.** *Fix $t_1 > 0$ with $p_1 := \pi([t_1, 2t_1]) > 0$, and assume $S \geq 1$. There is P_W such that, for every $\ell > 0$,*

$$\log\left(\frac{1}{\lambda_{\min}(K_W(\ell))}\right) \leq P_W + 4\log_+\left(\frac{\ell^2}{t_1}\right),$$

with $P_W = O(S \log n)$ whenever $g(y) \to 0$ as $y \to \infty$, in particular for every kernel considered here, and $P_W = O(\log(nS))$ under (A5)(ii). The implied constants depend only on g and t_1.

Remark 46 (Why the choice of B and S is not circular)**.** Pinning needs B of order $n_W P_W/\varepsilon_{\mathrm{p}}^2$, and the separation S must grow with B so that far-away blocks are negligible. Under (A5)(i), S/Λ need only grow polylogarithmically in B, so the bound $P_W = O(S \log n)$ is polylogarithmic in B up to the factor Λ. This is the only use of (A5)(i). Under (A5)(ii), S may grow polynomially, but P_W is only logarithmic in S. Either way the requirement on B is met by a polynomial. For a kernel with only polynomial decay and exponentially little mass of π near 0, P_W would grow polynomially in B and the argument would be circular; we know of no standard kernel of that kind.

Lemma 47 (Noise-free model)**.** *Let T_1 satisfy $\int_{t>T_1} t^2 \, \mathrm{d}\pi \leq m_2/2$. Assume*

$$v^\circ \leq \min\left\{\frac{v_1}{2}, \frac{m_1}{18 m_2}\right\}$$

and $\Lambda \geq 4T_1$. Then (H1) and (H2) hold with

$$\begin{aligned}
\Psi &= \log_+(1/\sigma_f^2) + 4\log_+(\ell^2/t_1), \\
\Gamma_0 &= \tfrac{1}{2} n_W (P_W + \log \bar{\lambda}_W) + \tfrac{1}{2}\bar{\lambda}_W \|\bar{\alpha}\|^2 + 1, \\
\mathfrak{b} &= B/32, \\
\Psi_0 &= 4c_4 + 20c_1 + 8,
\end{aligned}$$

where

$$c_1 = \log\left(\frac{4}{s_{\min}}\right) + 1, \quad s_{\min} = \min_g s_g, \quad c_4 = \log_+\left(\frac{18\Lambda}{t_1}\right).$$

Proof. (H1) follows from (2), $\lambda_{\min}(\Sigma_W(\theta)) = \sigma_f^2 \lambda_{\min}(K_W(\ell))$, and Lemma 45. For (H2), write $\chi_\sigma = \log_+(1/\sigma_f^2)$ and bound the left side from below using two disjoint sets of groups.

Small σ_f^2. Every $\kappa_g(\ell) \leq 4$, so if $\sigma_f^2 \leq 1$, then $-w_g \geq \chi_\sigma - \log(4/s_{\min})$. The group Sq-$aa'$ has the smallest target eigenvalue. Every within and pattern group has $\kappa_g(\Lambda) \geq F(v^\circ)$, every between group has $\kappa_g(\Lambda) \geq 1 \geq F(v^\circ)$, and

$$F(v^\circ) \geq \tfrac{1}{2} m_1 v^\circ \geq 9 m_2 v^{\circ 2} \geq \kappa_{\mathrm{Sq},aa'}(\Lambda)$$

by Lemma 7(b),(d) and $v^\circ \leq m_1/(18m_2)$. Since $s_g \geq e^{-2\delta}\lambda_g(\theta^\circ)$,

$$s_{\min} \geq \frac{1}{2}\lambda_{\mathrm{Sq},aa'}(\Lambda) \geq \frac{m_2}{4\Lambda^2}$$

by Lemma 7(d), since

$$\lambda_{\mathrm{Sq},aa'}(\Lambda) = \int (1 - e^{-tv^\circ})^2 \,\mathrm{d}\pi \quad \text{and} \quad \frac{1}{v^\circ} = \frac{\Lambda}{2} \geq T_1.$$

All groups except Sq-aa' and P_1-within have total dimension at least $5B$, and because $\psi(w) \geq \frac{1}{4}(|w| - 1)$, they contribute at least $\frac{5B}{8}(\chi_\sigma - c_1)$. If $\sigma_f^2 > 1$ or $\chi_\sigma < c_1$, this bound is negative and holds trivially, since $\psi \geq 0$.

Large ℓ. Suppose $\ell^2 \geq 18\Lambda$, so

$$v = \frac{2}{\ell^2} \leq \frac{v^\circ}{18} \leq \frac{m_1}{m_2}.$$

By Lemma 7,

$$\kappa_{P_1\text{-within}}(\ell) = F(v) \geq \frac{1}{2} m_1 v \quad \text{and} \quad \kappa_{\mathrm{Sq},aa'}(\ell) = \int (1 - e^{-tv})^2 \mathrm{d}\pi \leq m_2 v^2.$$

At the target, $\kappa_{\mathrm{Sq},aa'}(\Lambda) \geq \frac{1}{8} m_2 v^{\circ 2}$ and $\kappa_{P_1\text{-within}}(\Lambda) \leq m_1 v^\circ$, so with rounding,

$$\frac{s_{aa'}}{s_{P_1}} \geq \frac{m_2}{9 m_1} v^\circ.$$

Hence

$$|w_{aa'}| + |w_{P_1}| \geq w_{P_1} - w_{aa'} \geq \log\left(\frac{v^\circ}{18v}\right) = \log\left(\frac{\ell^2}{18\Lambda}\right) =: \chi_\ell,$$

and these two groups, each of dimension B, contribute at least $\frac{B}{8}(\chi_\ell - 2)$. If $\ell^2 < 18\Lambda$, they contribute at least $0 \geq \frac{B}{8}((\chi_\ell)_+ - 2)$, so in every case, they contribute at least $\frac{B}{8}((\chi_\ell)_+ - 2)$. Adding,

$$X \geq \frac{B}{8}(\chi_\sigma + (\chi_\ell)_+) - \frac{5B}{8} c_1 - \frac{B}{4}.$$

Since $\log_+(\ell^2/t_1) \leq (\chi_\ell)_+ + c_4$, we have $\Psi \leq 4(\chi_\sigma + (\chi_\ell)_+) + 4c_4$, and therefore

$$X \geq \frac{B}{32}(\Psi - 4c_4) - \frac{5B}{8} c_1 - \frac{B}{4} = \frac{B}{32}(\Psi - 4c_4 - 20c_1 - 8). \qquad \square$$

The rest of this appendix proves Lemma 45.

Lemma 48 (Truncated Gaussian moments)**.** *Let $\nu_1, \ldots, \nu_J$ be finitely supported signed measures on $\mathbb{R}^D$, and*

$$E_\tau[j, j'] = \iint e^{-\tau \|x-y\|^2} \mathrm{d}\nu_j(x) \,\mathrm{d}\nu_{j'}(y).$$

Let $\mathcal{M}'$ be a set of at least J distinct monomials of degree at most 4, and

$$\Phi_\tau[j, \alpha] = \int e^{-\tau \|x\|^2} x^\alpha \,\mathrm{d}\nu_j(x)$$

for $\alpha \in \mathcal{M}'$. Then

$$\lambda_{\min}(E_\tau) \geq \frac{2}{3} \min\{1, \tau^4\} \, \sigma_{\min}(\Phi_\tau)^2,$$

where $\sigma_{\min}$ is the J-th largest singular value.

Proof. Expand

$$e^{-\tau\|x-y\|^2} = e^{-\tau\|x\|^2} e^{-\tau\|y\|^2} \sum_{k\geq 0} \frac{(2\tau)^k}{k!} (x\cdot y)^k, \quad (x\cdot y)^k = \langle x^{\otimes k}, y^{\otimes k}\rangle.$$

For $c \in \mathbb{R}^J$, let

$$\nu_c = \sum_j c_j \nu_j, \quad T_k = \int e^{-\tau\|x\|^2} x^{\otimes k} \, \mathrm{d}\nu_c(x).$$

Then

$$c^\top E_\tau c = \sum_{k\geq 0} \frac{(2\tau)^k}{k!} \|T_k\|^2.$$

Every monomial x^α of degree k equals at least one entry of $x^{\otimes k}$, so

$$\|T_k\|^2 \geq \sum_{\alpha\in\mathcal{M}', |\alpha|=k} (\Phi_\tau^\top c)_\alpha^2.$$

Keeping only $k \leq 4$ and using $\min_{k\leq 4}(2\tau)^k/k! \geq \frac{2}{3}\min(1,\tau^4)$ gives

$$c^\top E_\tau c \geq \frac{2}{3}\min\{1,\tau^4\}\|\Phi_\tau^\top c\|^2 \geq \frac{2}{3}\min\{1,\tau^4\}\sigma_{\min}(\Phi_\tau)^2\|c\|^2. \qquad \square$$

Lemma 49 (Unisolvency of the gadget)**.** *Let $\mathcal{M}$ consist of the monomials 1; x_i^r for $i \in [d]$ and $r = 1,\dots,4$; and $x_j^r x_k^{r'}$ for $\{j,k\} \in E$ and $r, r' \in \{1,2\}$. Then $|\mathcal{M}| = n_0$, and the evaluation matrix $\Phi_G = [p^\alpha]_{p\in P, \alpha\in\mathcal{M}}$ is invertible with*

$$\|\Phi_G^{-1}\|_2 \leq 65\sqrt{(d+1)(m+1)} =: \sigma_G^{-1}.$$

Proof. Order the rows as the origin, the axis points grouped by coordinate, and the edge points grouped by edge, and the columns as the constant, the axis monomials grouped by coordinate, and the edge monomials grouped by edge. The origin vanishes on every nonconstant monomial. An axis point te_i vanishes on every edge monomial and on every axis monomial of another coordinate. An edge point $ae_j + be_k$ vanishes on every edge monomial of another edge, because such a monomial involves a coordinate outside $\{j,k\}$. Hence Φ_G is block lower triangular,

$$\Phi_G = \begin{bmatrix} 1 & 0 & 0 \\ B_1 & I_d \otimes V & 0 \\ C_1 & C_2 & I_m \otimes H \end{bmatrix}, \quad V = [t^r]_{t\in\{\pm 1, \pm\frac{1}{2}\}, r=1,\dots,4}, \quad H = [a^r b^{r'}]_{a,b\in\{\pm 1\}, r,r'\in\{1,2\}}.$$

Since $a^2 = b^2 = 1$, the columns of H are ab, a, b and 1, which are orthogonal sign vectors, so $H^\top H = 4I$ and $\|H^{-1}\| = \frac{1}{2}$. A direct computation gives $\|V^{-1}\| \approx 5.399 \leq 5.4$. The off-diagonal blocks have entries in $\{0, \pm 1\}$: $\|B_1\| = 2\sqrt{d}$ and $\|C_1\| = 2\sqrt{m}$; each edge row has eight nonzero entries in C_2, so $\|C_2\| \leq \|C_2\|_F = \sqrt{32m}$. Inverting the block triangular matrix and bounding each block of the inverse,

$$\|\Phi_G^{-1}\| \leq 1.5 + 5.4 + 10.8\sqrt{d} + 16.3\sqrt{m} + 30.6\sqrt{md} \leq 65\sqrt{(d+1)(m+1)}. \qquad \square$$

Corollary 50 (Gadget Gram matrix)**.** *Fix $t_1 > 0$ with $p_1 := \pi([t_1, 2t_1]) > 0$, and let $K_{GG}(\ell)$ be the Gram matrix of P at length scale ℓ. Then for every $\ell > 0$,*

$$\lambda_{\min}(K_{GG}(\ell)) \geq c_G p_1 \min\left\{1, \left(\frac{t_1}{\ell^2}\right)^4\right\}, \qquad c_G := \min\left\{\frac{1}{2}, \frac{2\sigma_G^2}{3(2n_0)^{16}}\right\}.$$

Proof. Let $E_\tau = [e^{-\tau\|p-p'\|^2}]_{p,p'\in P}$, so $K_{GG}(\ell) = \int E_\tau \, \mathrm{d}\pi_\ell(\tau)$. Take $\nu_p = \delta_p$ in Lemma 48. Then $\Phi_\tau = \mathrm{diag}(e^{-\tau\|p\|^2})\Phi_G$ and $\|p\|^2 \le 2$, so $\sigma_{\min}(\Phi_\tau) \ge e^{-2\tau}\sigma_G$. For $\tau \le \tau_G := 4\log(2n_0)$ this gives

$$\lambda_{\min}(E_\tau) \ge \frac{2}{3}\min\{1,\tau^4\}(2n_0)^{-16}\sigma_G^2.$$

For $\tau \ge \tau_G$, distinct gadget points are at squared distance at least $\frac{1}{4}$, so every off-diagonal entry is at most $e^{-\tau/4} \le 1/(2n_0)$, and Gershgorin gives $\lambda_{\min}(E_\tau) \ge \frac{1}{2}$. In both cases $\lambda_{\min}(E_\tau) \ge c_G \min\{1,\tau^4\}$. Since $c^\top K_{GG} c = \int c^\top E_\tau c \, \mathrm{d}\pi_\ell$, restricting to $\tau \in [t_1/\ell^2, 2t_1/\ell^2]$, which has π_ℓ-mass p_1, proves the claim. □

Proof of Lemma 45. Translate every non-main coordinate by $+1$. This changes no kernel value. Afterwards gadget points have zero non-main coordinates, and every point of family f has squared norm $4S + \varrho_f \le 4S+4$. Let $\nu_p = \delta_p$ for $p \in P$ and $\nu_f = n_f^{-1/2}\sum_{x\in f}\delta_x$. Then $K_W(\ell) = \int E_\tau \, \mathrm{d}\pi_\ell(\tau)$ with E_τ as in Lemma 48. We bound $\lambda_{\min}(E_\tau)$ for small and large τ and then integrate.

Small scales. Fix one hub coordinate $c_f^{\mathrm{hub}} \in S_{b_f}$ in one block b_f of each family, and apply Lemma 48 with the monomials $\mathcal{M} \cup \{x_{c_f^{\mathrm{hub}}}\}_f$. Then $\Phi_\tau = D_\tau\tilde{\Phi}$, where D_τ is diagonal with entries $e^{-\tau\|p\|^2}$ for gadget rows and $e^{-\tau(4S+\varrho_f)}$ for family rows, and

$$\tilde{\Phi} = \begin{bmatrix} \Phi_G & 0 \\ Z_1 & Z_2 \end{bmatrix}$$

does not depend on τ. The zero block appears because gadget points have zero hub coordinates. Points of a family have zero main coordinates, so a family row vanishes on the nonconstant monomials of $\mathcal{M}$; its constant entry is $\sqrt{n_f}$, so $\|Z_1\| \le \sqrt{n}$. Its entry for $x_{c_{f'}^{\mathrm{hub}}}$ is $2k_f/\sqrt{n_f}$ if $f' = f$ and 0 otherwise, so Z_2 is diagonal with $\|Z_2^{-1}\| \le \sqrt{n_{\max}}/4$. Hence

$$\|\tilde{\Phi}^{-1}\| \le \sigma_G^{-1}(1 + \|Z_2^{-1}\|\|Z_1\|) + \|Z_2^{-1}\| \le \sigma_G^{-1}(1+n) + \sqrt{n} =: \sigma_W^{-1}, \quad \sigma_{\min}(\Phi_\tau) \ge e^{-\tau(4S+4)}\sigma_W.$$

Lemma 48 gives

$$\lambda_{\min}(E_\tau) \ge \frac{2}{3}\min\{1,\tau^4\}e^{-2\tau(4S+4)}\sigma_W^2.$$

Large scales. Let $\tau_W = 4\log(8n_0 n_{\max})$. The diagonal entries of E_τ are at least 1. Off the diagonal, gadget–gadget entries are at most $e^{-\tau/4}$, gadget–family entries at most $\sqrt{n_f}e^{-4S\tau}$, and family–family entries at most $\sqrt{n_f n_{f'}}e^{-8S\tau}$. For $\tau \ge \tau_W$ and $S \ge 1$, every row's off-diagonal sum is at most $\frac{1}{8} + n_0 n_{\max}e^{-4\tau} \le \frac{1}{2}$, and Gershgorin gives $\lambda_{\min}(E_\tau) \ge \frac{1}{2}$. Together, for every $\tau > 0$,

$$\lambda_{\min}(E_\tau) \ge f_W(\tau) := \min\left\{\frac{1}{2}, \frac{2}{3}\min\{1,\tau^4\}e^{-2\tau(4S+4)}\sigma_W^2\right\}, \qquad \lambda_{\min}(K_W(\ell)) \ge \int f_W \, \mathrm{d}\pi_\ell.$$

Integration. Since $g(y) \to 0$ as $y \to \infty$, π has no atom at 0; this is all we use here, and (A2) is one sufficient condition. Hence there is $t_{\mathrm{med}} > 0$ with $\pi([t_{\mathrm{med}}, \infty)) \ge \frac{1}{2}$. Put $\ell_{\min}^2 = t_{\mathrm{med}}/\tau_W$, and write $t = \tau\ell^2$, so that $t \sim \pi$.

- If $\ell^2 \le \ell_{\min}^2$, then
$$\pi_\ell(\tau \ge \tau_W) \ge \pi(t \ge t_{\mathrm{med}}) \ge \frac{1}{2},$$
so $\lambda_{\min} \ge \frac{1}{4}$.

- If $\ell^2 \geq 32St_1$, use $t \in [t_1, 2t_1]$. Then

$$2\tau(4S+4) \leq 4t_1 \cdot \frac{8S}{\ell^2} \leq 1, \quad \tau \geq \frac{t_1}{\ell^2},$$

so

$$\lambda_{\min} \geq \frac{2}{3e} p_1 \sigma_W^2 \min\left\{1, \left(\frac{t_1}{\ell^2}\right)^4\right\}.$$

This is the only case that produces the term $4\log_+(\ell^2/t_1)$.

- If $\ell^2_{\min} < \ell^2 < 32St_1$, use $t \in [t_1, 2t_1]$. Then $\tau \leq 2t_1\tau_W/t_{\mathrm{med}}$, so

$$2\tau(4S+4) \leq \frac{32St_1\tau_W}{t_{\mathrm{med}}} = O(S\log n),$$

and $\min\{1, \tau^4\} \geq (32S)^{-4}$.

- Under (A5)(ii), the previous case can be refined. If $\ell^2_{\min} < \ell^2 \leq 20Su_0$, use

$$t \in \left[\frac{\ell^2}{20S}, \frac{\ell^2}{10S}\right]; \quad \text{i.e.,} \quad \tau \in \left[\frac{1}{20S}, \frac{1}{10S}\right].$$

Then $2\tau(4S+4) \leq 1.6$ and $\tau^4 \geq (20S)^{-4}$, and the π-mass of the window is at least

$$c_\pi \left(\frac{\ell^2_{\min}}{20S}\right)^{b_\pi}.$$

- Under (A5)(ii), if $20Su_0 < \ell^2 < 32St_1$, use $t \in [t_1, 2t_1]$. Then $2\tau(4S+4) \leq 1.6\, t_1/u_0$ and $\min\{1, \tau^4\} \geq (32S)^{-4}$.

In every case, $\log(1/\lambda_{\min})$ is at most $4\log_+(\ell^2/t_1)$ plus the logarithm of the reciprocal of a product of the displayed factors, p_1 and σ_W^2. Since $\log(1/\sigma_W) = O(\log n)$, this is $O(S\log n)$ in general, and $O(\log(nS))$ under (A5)(ii) by the two refined cases. □

C The group part of the posterior mean

We prove Lemma 21(d): $|\mu_{\mathrm{grp}}| \leq 1.1$ on $[-1,1]^D$ for every $\theta \in \mathcal{W}_\eta$. Fix x. As in the proof of Lemma 21(b), at most one block, the *near block*, has $D_b(x) < 2S$, and every point of every other block is at squared distance at least S from x, so its kernel value is at most $g(S/\ell^2) \leq g(S/(2\Lambda))$. Two points of a pair, or two points of a square that differ in one sign, differ in exactly two private coordinates, where one point has $-1+h$ and the other -1. A private coordinate x_i of x lies in $[-1,1]$, so each such coordinate changes $(x_i - z_i)^2$ by $2h(x_i+1) - h^2 \in [-h^2, 4h - h^2]$, and the squared distances from x to the two points differ by at most $4h$. Since g is m_1-Lipschitz (Lemma 7(a)), their kernel values differ by at most $4hm_1/\ell^2$. Opposite corners of a square can differ by up to $8h$, but no bound below uses them. Throughout, $\ell^2 \geq 0.98\Lambda \geq 12500m_1$ and $\ell^2 \leq 1.02\Lambda$.

Far blocks. The coefficient of a far observation in μ_{grp} is $\sigma_f^2 \sum_g (y_g)_{\mathrm{obs}}/\lambda_g(\theta)$, a sum over at most four groups. Each $|(y_g)_{\mathrm{obs}}| \leq 1.01$ (all group magnitudes are at most 1.01), and every $\lambda_g(\theta) \geq \sigma_f^2 r \kappa_g(\ell) \geq 0.98\, m_2/(2.1\Lambda^2)$, the smallest being the square pattern aa' (Lemma 7(d) with $v = 2/\ell^2$). So each far coefficient is at most $9\Lambda^2/m_2 = 2.25M$ in the noise-free model; in the noisy model the group eigenvalues are at least $\sigma_n^2 \geq 0.98$, so each far coefficient is at most 5. In both cases the far blocks contribute at most $3nM\, g(S/(2\Lambda)) \leq 0.003$ by (3).

Near block, noisy model. The near block belongs to a pair family P_h.

- *Within.* The contribution is

$$\frac{\sigma_f^2 r\zeta_h[k_\ell(x,p_{b,2})-k_\ell(x,p_{b,1})]}{\lambda_{P_h\text{-within}}(\theta)}.$$

  Here,

$$\lambda_{P_h\text{-within}} \geq r\sigma_f^2 F(2h^2/\ell^2) \geq r\sigma_f^2 \cdot \frac{m_1 h^2}{\ell^2}$$

  by Lemma 7(b), since $2h^2/\ell^2 \leq m_1/m_2$, and

$$\zeta_h^2 = \frac{rF(2h^2/\Lambda)+1}{2r} \leq \frac{m_1 h^2}{\Lambda} + \frac{1}{2r} \leq \frac{4m_1 h^2}{\ell^2},$$

  using $r \geq \Lambda/m_1$. So the contribution is at most $4\zeta_h/h \leq 8\sqrt{m_1}/\ell \leq 0.072$.

- *Between.* The contribution is

$$\frac{\sigma_f^2 r m_b[k_\ell(x,p_{b,1})+k_\ell(x,p_{b,2})]}{\lambda_{P_h\text{-between}}(\theta)}.$$

  Here, $\lambda_{P_h\text{-between}} \geq 1.99 r\sigma_f^2$, because $g(8/\ell^2) \geq 1 - 8m_1/\ell^2 \geq 0.999$ and $g_\ell(8S) \leq 0.0001$. In addition, $|m_b| = \xi_h \leq 1.0001$, since

$$\xi_h^2 = \frac{B-1}{2rB}\,\lambda_{P_h\text{-between}}(\theta^\circ) \leq \frac{2r+1}{2r} \leq 1.0001.$$

  The contribution is therefore at most $2 \cdot 1.0001/1.99 \leq 1.006$.

In conclusion, the near block contributes at most 1.08 in absolute value.

Near block, noise-free model. Here $r = 1$, $\sigma_n = 0$, and μ carries no factor σ_f^2; the group coefficients are $(y_g)_{\text{obs}}/\kappa_g(\ell)$.

- *Pair, within.* The contribution is

$$\frac{\zeta_h[k_\ell(x,p_{b,2})-k_\ell(x,p_{b,1})]}{\kappa_{P_h\text{-within}}(\ell)}.$$

  By Lemma 7,

$$\kappa_{P_h\text{-within}}(\ell) = F(2h^2/\ell^2) \geq m_1 h^2/\ell^2, \quad \zeta_h^2 \leq 1.01 m_1 h^2/\Lambda,$$

  so the contribution is at most $4.1\sqrt{m_1/\Lambda}$.

- *Square, patterns a and a'.* The contribution is

$$\zeta_a \sum_{a,a'} \frac{a\,k_\ell(x,p_{aa'})}{\kappa_{\text{Sq},a}(\ell)}.$$

  Pairs differing in a have kernel values within $4m_1/\ell^2$, so the sum is at most $8m_1/\ell^2$, while

$$\kappa_{\text{Sq},a}(\ell) = F(4/\ell^2) \geq \frac{2m_1}{\ell^2}, \quad \zeta_a^2 \leq \frac{1.01 m_1}{\Lambda}.$$

  Each is at most $4.1\sqrt{m_1/\Lambda}$.

- *Square, pattern aa'.* The squared distance from x to point (a, a') has the form $D' + A_a + A'_{a'}$ with $|A_+ - A_-|, |A'_+ - A'_-| \le 4$, so $\sum_{a,a'} aa'\, k_\ell(x, p_{aa'})$ is a mixed second difference of g, at most $m_2 \cdot 16/\ell^4$. Also, $\kappa_{\mathrm{Sq},aa'}(\ell) \ge m_2/(2\ell^4)$ by Lemma 7(d), and $\zeta_{aa'} \le 1.01\sqrt{m_2}/\Lambda$, so the contribution is at most $32.4\sqrt{m_2}/\Lambda$.

- *Between.* The contribution is at most

$$\frac{|m_b|\, k_f}{\kappa_{f\text{-between}}(\ell)},$$

  with $|m_b| = \xi_f \le 1$. Because $g(u) \ge 1 - m_1 u$ and $\ell^2 \ge 12500 m_1$, $\kappa_{f\text{-between}}(\ell) \ge 1.99$ for pairs and ≥ 3.99 for squares, so the contribution is at most 1.006.

With $\Lambda \ge 12800 m_1 + 10^4\sqrt{m_2}$, the near block contributes at most

$$1.006 + 8.2\sqrt{\frac{m_1}{\Lambda}} + 32.4\frac{\sqrt{m_2}}{\Lambda} \le 1.09.$$

In both models, therefore, $|\mu_{\mathrm{grp}}| \le 1.09 + 0.003 \le 1.1$. □